Full-phase complex Hamiltonian learning with conservative certification and dissipative magnetic feedback

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Abstract

We connect a trained complex Hamiltonian model to conservative stability and dissipative feedback for a planar magnetic rod with prescribed configuration-dependent inertia. The certified checkpoint has two complex hidden layers and 736 real parameters, learns a full-phase energy correction from fourteen COMSOL trajectories, and uses a separately fitted positive-semidefinite damping head. Signed observation residuals, label fidelity and verified analytic remainders bound its first- through third-order energy-error derivatives. A separate 16,904-anchor numerical-reference experiment validates a finite-output acceptance route for this same checkpoint. These bounds enter a completed conservative-family certificate, including the specified reference endpoint. A physical-energy ledger and validated initial-value tubes separately prove an energy cap throughout 12,567 held magnetic commands covering one reference period. Additional real and complex fits use a common output scale and soft penalties on the physical second- and fourth-order energy jets. On the reused full-phase anchor record, the new complex output has approximately 53.2% lower vector RMS error than the fourth-order prior and 5.0% lower error than the new real output, while its velocity-channel error is larger. These single-seed exploratory comparisons establish neither general complex superiority nor full-error, KAM or control certificates for the new outputs. The population-risk route for the full network class remains conditional. All mathematical and control certificates concern their stated finite-dimensional models and declared numerical viscosity.

Keywords

complex Hamiltonian learning, magnetic soft robots, observation-based error bounds, a posteriori KAM theory, sampled feedback

1 Introduction

Magnetic actuation couples remotely generated fields to magnetized bodies, allowing mechanical input to be delivered without a transmission attached to each moving segment. The resulting motion depends jointly on the actuation field, the body's magnetic properties and its deformation (Abbott et al. 2020). Flexible magnetic robots illustrate how distributed magnetization can support substantial changes of shape and steering in confined environments (Kim et al. 2019). A predictive model is useful for relating an applied command to this evolving configuration. When that model is learned from data, feedback also requires a quantitative account of the effects of its remaining error. A close trajectory fit alone does not specify the domain, time interval or energy margin over which a subsequent stability claim is valid.

Rod and beam mechanics provide a starting point for this analysis. Geometrically nonlinear rod descriptions distinguish the centerline, section orientation, elastic response and applied loads (Antman 2005; Simo 1985). Reduction to a small number of generalized coordinates makes identification and control accessible while preserving selected geometric dependencies. Here a planar magnetic-rod chart provides a finite mechanical reference with elastic energy, magnetic loading and distributed viscous loss. This reference fixes the quantities to which the neural approximation and its certificates refer. The development links mechanical reduction, complex energy learning, conservative verification and dissipative feedback through one declared canonical model.

An important feature of the reduction is its configuration-dependent inertia. The canonical momentum is obtained from velocity through the positive metric, and the kinetic

energy depends on configuration as well as momentum. Its configuration derivative therefore enters the momentum equation. Retaining this dependence provides a consistent conversion between the measured motion, the scalar energy and its canonical field. It also distinguishes a prescribed mechanical prior from the functions that data must determine. We retain the known kinetic graph and a fourth-order potential, then allow a neural correction to depend on all four canonical coordinates. This permits momentum-dependent model error to enter both identification and verification explicitly.

Hamiltonian neural networks learn a scalar energy and differentiate it to obtain a Hamiltonian vector field (Greydanus et al. 2019). The scalar representation links the components of the predicted dynamics and imposes the conservative differential structure algebraically. A mechanical prior can further specify an inertia or an elastic component while leaving a smaller correction to be learned. For the present full-phase correction, this construction retains a real learned energy even though the hidden features and weights are complex. The learned velocity is the momentum derivative of that complete scalar. It contains the network's momentum derivative in addition to the velocity associated with the prescribed kinetic energy.

Dissipation and input have complementary roles. Dissipative SymODEN encodes a port-Hamiltonian computation graph containing energy loss and control (Zhong et al. 2020). Port-Hamiltonian neural networks also separate a stationary energy from dissipative and driving terms in learning non-autonomous dynamics (Desai et al. 2021). We use a configuration-dependent factor decoder for a real positive-semidefinite dissipation matrix. Its quadratic Rayleigh function gives a dissipative force through a velocity derivative. The canonical decoder combines that force with the full learned Hamiltonian velocity and an explicit magnetic port. This construction yields a checkable power identity and specifies how a separately fitted matrix head can be paired with the conservative energy. The pair's mathematical identity is maintained when its errors enter a reference-plant controller.

Complex networks offer a concrete parameterization for these scalar and matrix functions. Trabelsi et al. (2018) develop building blocks for deep complex learning, including complex affine operations and their real implementation. The conservative model used here has two literal complex-tanh hidden layers, independent affine coefficients for the complex input and its conjugate on real states, and a real scalar readout. Evenization and an energy gauge fix its symmetry and value at the origin. For analytic verification, the four real canonical coordinates are extended independently to complex arguments. The real readout then has a coefficient-conjugate analytic continuation. Avoidance of the tanh poles becomes a candidate-specific mathematical condition on that continuation.

The use of complex arithmetic also raises an empirical question about representation. Zhang et al. (2022) study approximation and optimization properties of specified complex-reaction network families. Those results motivate examining network structure and training conditions when comparing real and complex models. Our finite-width complex-tanh model, its constrained parameterization and its Adam objective are evaluated through their actual returned functions. We compare real and complex architectures both with and without a prescribed quadratic Hamiltonian, and separately examine the effect of objective and output scaling. This separates the practical choice of a representation from the canonical and power identities, which follow from the scalar and decoder construction.

Nonzero approximation error is central to the stability question. Chen et al. (2022) connect Hamiltonian neural networks with nonzero training loss to KAM theory through statistical learning and regularity estimates. Their analysis makes the sampling distribution and control of the relevant function class substantive ingredients. Classical KAM theory describes persistence of suitable quasiperiodic invariant tori under small Hamiltonian perturbations, with arithmetic and nondegeneracy conditions (Pöschel 2001). For a computed mechanical model, an a posteriori theorem starts from a proposed torus and checks its invariance defect, analytic domain and geometry (Haro and Luque 2019). Learning error can contribute to these estimates, but the target torus and its complete defect remain additional objects to be verified.

Trajectory observations constrain a different set of quantities from those analytic hypotheses. Successive observations share prescribed motions and initial conditions. Weak momentum balances constrain integrals along those motions, whereas an analytic stability proof needs derivatives on a whole state domain. Configuration-gradient observations at zero momentum give useful potential information but do not determine arbitrary momentum dependence. We therefore combine rest-gradient information with momentum-gradient residuals at observation-chart anchors. Because the kinetic graph is shared, the reference correction has an exactly zero momentum derivative at those anchors. This structural identity provides a full-phase observation channel without using exported native momentum derivatives or reference forces as training labels.

The realized observation design is verified through finite-dimensional representatives of the actual network output. For the degree-six representative, all nonconstant even-total-degree monomials in the four canonical variables give 129 coordinates. Fifteen configuration-only coordinates and 114 mixed coordinates are recovered with validated inverse bounds. The neural function remains the deployed function; the representative serves to transport its signed observation residual and its actual remainder. Observation-point tails, target-domain tails, label fidelity and the inverse correction enter the same inequality. This yields complete mixed-derivative bounds rather than inferring full-phase accuracy from a configuration slice.

A prospective sampling result complements this realized-design calculation. Independent anchors under a specified importance law can give a common conditioning event for the derivative design. On that event, every candidate with sound individual residual and remainder certificates satisfies the corresponding output-error implication. Consequently an output may be selected using the same observations without a union bound over all returned network parameters. The probability statement controls accepting an output whose asserted guarantee is false. Its assumptions remain explicit, including the independent sampling law and common fidelity event. The prescribed fourteen trajectories support the native deterministic certificate. A separate experiment uses fresh canonical reference-oracle anchors and verifies its realized importance-weighted design, as reported below.

Time discretization supplies another structural component. Symplectic integration preserves the symplectic form of a conservative Hamiltonian flow; implicit midpoint is a standard example (Hairer 2010). We formulate that method for the nonseparable mechanical energy and its complex-coordinate representation. Local solvability and global approximation estimates retain the contribution of a finite nonlinear-solver residual. The numerical approximation, continuous-flow KAM calculation and recorded COMSOL observations thus have their own specified roles. This allows the discretization analysis to support the same mechanical definitions without substituting a discrete-map property for a continuous invariant-torus certificate.

The checkpoint used for conservative and feedback certification is a 736-real-coordinate complex energy correction with an already fitted 42-coordinate PSD matrix head held fixed. It uses fourteen distributed-viscosity COMSOL trajectories and 400 full-batch Adam updates. Selection by the training objective gives update 385 before the reused rest labels are evaluated. Its complete four-dimensional observation-error certificate bounds every mixed derivative through order three on the stated complex rectangle. We write $\widehat{H}_0 = H_4 + N_0$ for this precision-centered GC checkpoint at θ_0. The three factorial GC outputs and the additional jet-regularized outputs retain their own parameter identities. A separate computer-assisted calculation proves an analytic invariant two-torus for this same learned Hamiltonian. The observation-derived error bounds also enter a completed conservative-family certificate. Uniform domains, the actual varying mean torsion and a complete whole-strip defect for a degree-ten response satisfy all eight a posteriori inequalities. For each real family parameter, these bounds certify a real analytic embedded torus with the same exact rotation vector, including at the prescribed conservative reference endpoint.

For magnetic feedback, the difference between the learned and reference velocities contributes to both the magnetic work allowance and the dissipation credit. We retain this momentum-gradient error when constructing the held-command filter. A rational energy ledger and validated initial-value tubes then establish a strict physical-reference energy cap throughout 12,567 holds covering a complete reference period, with a continuous-time bound on the output error. The result concerns the specified numerical mechanical reference and distributed viscosity. It supplies an entire-hold interpretation of the sampled decisions while keeping reference calibration and hardware interfaces outside that numerical assertion.

The comparisons also expose the limits of the learned correction. The real/complex by quadratic-prior experiment retains five paired contrasts over three common-data optimization seeds. It favors the complex architecture under its original budget, but its absolute rest-gradient errors exceed the shared fourth-order prior's error. A precision-centered prior-assisted comparison reverses the real–complex ranking: the real output is more accurate, and both selected outputs remain worse than that prior on the reused labels. The real dissipation head also performs better on the reported matrix identification metrics. These results make initialization, scaling, architecture and the physical prior part of the interpretation of a representation comparison.

An additional experiment addresses the known low-order structure of the reference correction. It penalizes the physical gradients of the second- and fourth-order neural energy jets through exact uniform-box Grams, while retaining both ordinary full-phase architectures. Two single-seed fits share an actual output scale, the initial function H_4, a projected-Adam update budget, and training-only selection. Both returned functions improve on H_4 on the reported native and

reused full-phase anchor records. The new complex output has smaller total anchor error than the new real output but a larger anchor velocity error. These are post-hoc numerical-reference comparisons, with no claim of an independent test or causal complex superiority. Their new parameter identities are kept separate from the original KAM and magnetic-feedback certificates.

The contribution is a quantitative connection between an actually trained full-phase complex energy, its observation-error certificate, the conservative learned–reference family's invariant tori and a separate dissipative physical-reference safety certificate. The analysis keeps the network, canonical chart, finite mechanical graph and decoder consistent across these stages. The native observations and the separate independent-anchor experiment provide two verified error routes for the same fixed output. The latter uses a finite-output common event; a full-network population-risk guarantee retains its additional class-wide assumptions. The resulting framework allows structural validity, approximation accuracy and downstream mathematical acceptance to be assessed for the same returned functions.

The new complex output also has its own verified analytic continuation on the original target and its collar. This supplies one regularity condition, rather than a complete error, statistical, torus or control certificate for that output. A finite-polynomial population-risk bridge further makes the real-domain and complex-domain remainder requirements explicit; its present numerical application remains specific to the original θ_0.

2 Mechanical reduction and complex symplectic learning

2.1 Reduced rod mechanics and physical normalization

The physical reference is a planar, inextensible and unshearable Kirchhoff rod with a fixed base and a magnetic tip moment. Two prescribed shape functions supply the dimensionless coordinates $q \in \mathbb{R}^2$. Writing $\beta_i(s) = f_i'(s)$, its angular and centreline descriptions are

$$\theta(s,q) = \beta(s)^{\mathsf{T}} q, \qquad r(s,q) = \int_0^s \begin{pmatrix} \sin\theta(\xi,q) \\ \cos\theta(\xi,q) \end{pmatrix} \mathrm{d}\xi. \tag{1}$$

The spatial components are transverse x and axial z. The trigonometric tangent retains finite rotations, while the prescribed two-coordinate chart restricts the admissible shapes. No axial stretching or independent shear variable is added. The construction follows rod kinematics Antman (2005); Simo (1985); all conclusions below concern its specified finite reduction.

The cross-section follows the local tangent, so its angular velocity is linear in the coordinate velocities even when the centreline undergoes finite rotation. This accounts for the constant rotary contribution below. The bending constitutive law remains quadratic in curvature within this chart. Gravity, field gradients and additional body-force terms are absent from the reference equations. The free end receives the magnetic moment; its rotation is not prescribed as an essential boundary condition.

2.1.1 The fixed quadrature and inertia The shape functions are piecewise cubic on 32 cells of the 0.08 m rod. At a material location, the configuration Jacobian is

$$J(s,q) = \partial_q r(s,q) = \int_0^s \begin{pmatrix} \cos\theta(\xi,q) \\ -\sin\theta(\xi,q) \end{pmatrix} \beta(\xi)^{\mathsf{T}} \,\mathrm{d}\xi. \tag{2}$$

The implemented $J_{ek}(q)$ uses eight-point Gauss quadrature: completed cells contribute their full outer sums, and the current cell contributes a nested eight-point sum up to its outer node. With cell length ℓ_e and positive stored weights w_k, define

$$\begin{aligned} D(q) &= \sum_{e,k} \ell_e w_k J_{ek}(q)^{\mathsf{T}} J_{ek}(q), \\ G(q) &= f_m D(q) + K_r = S(q) S(q)^{\mathsf{T}}. \end{aligned} \tag{3}$$

Here K_r is the constant normalized rotary-inertia matrix, and $f_m > 0$ is the stored translational factor, nominally $\omega_0^2 \rho A_c / E_0$. In the mechanical and dissipation formulas, S denotes the lower Cholesky factor, not a further modal scaling. The known G is positive definite on the certified configuration box $|q_i| \le 1/10$, and S has strictly positive diagonal. The Gauss nodes, weights and mechanical coefficients are frozen binary64 values. Their exact stored values define the analytic reference; freshly evaluating a continuum integral or a parameter product would define a different reference. In particular, the stored f_m is retained in all identities involving $G - K_r$.

Translational kinetic energy is assembled from the velocities of material points, using their configuration Jacobians before the Gram products are formed. Consequently its inertia depends on configuration even though the two shape functions themselves are fixed. The same metric is used when constructing native trajectories, converting observed velocity to momentum, evaluating the learned Hamiltonian and verifying its conservative flow. The required derivatives differentiate this configuration dependence.

2.1.2 Canonical energy and dimensional quantities Let t denote SI time and set $\tau = \omega_0 t$, with $\omega_0 \simeq 11.417287282852458\,\mathrm{s}^{-1}$ and $E_0 = \mu B_0 \simeq 6.0 \times 10^{-7}\,\mathrm{J}$. Here μ is the tip magnetic dipole magnitude and B_0 the constant bias field. These scales and the shape functions are fixed before learning. A prime denotes $\mathrm{d}/\mathrm{d}\tau$, so $v = q' = \dot{q}/\omega_0$. Dividing physical energy by E_0 gives

$$\begin{aligned} p &= G(q)v, \\ H_*(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}G(q)^{-1}p + V_*(q), \\ V_*(q) &= \tfrac{1}{2}q^{\mathsf{T}}Bq + 1 - \cos(d^{\mathsf{T}}q). \end{aligned} \tag{4}$$

The fixed $B \succ 0$ is the normalized bending matrix, and the fixed vector d maps the chosen coordinates to tip rotation. Both are the exported coefficients of this chart. Thus the full bias-field cosine and the configuration-dependent inertia are retained. The momentum p is dimensionless canonical momentum, not SI velocity; if P_{SI} is the momentum conjugate to the same dimensionless q, then $p = (\omega_0/E_0)P_{\mathrm{SI}}$.

Equivalently, the dimensionless Lagrangian is $L_*(q,v) = v^{\mathsf{T}}G(q)v/2 - V_*(q)$, and the momentum definition in (4) is its Legendre transform. In particular, configuration derivatives of the kinetic energy contribute to the momentum equation. They cannot be replaced by the force of a constant-mass oscillator. The prescribed mechanical graph defines both the numerical reference and the observation chart. The learned scalar adds a correction on the full position–momentum space. Its momentum derivative changes the learned velocity; the reference identity $v = G^{-1}p$ is not imposed on that learned velocity.

Distributed viscous drag supplies the physical Rayleigh coefficient $C_{\mathrm{phys}} = c_{\mathrm{line}}D(q)$, whose units are J s: J has units m, D has units m^3, and c_{line} has units $\mathrm{N\,s/m^2}$. Its normalized damping is

$$\begin{aligned} R_*(q) &= \frac{\omega_0 c_{\mathrm{line}}}{E_0}D(q) \\ &= a\{G(q) - K_r\}, \\ a &= \frac{\omega_0 c_{\mathrm{line}}}{E_0 f_m}, \\ \mathcal{D}_*(q,v) &= \tfrac{1}{2}v^{\mathsf{T}}R_*(q)v. \end{aligned} \tag{5}$$

The declared $c_{\mathrm{line}} = 0.0006911873093836746\,\mathrm{N\,s/m^2}$ is a numerical-study assumption inherited from a nominal small-slope damping estimate, not a measured material calibration. The present reference sets tip drag $c_{\mathrm{tip}} = 0$ and additional mass-proportional drag $\gamma_{\mathrm{add}} = 0$. Its damping is the matrix in (5); no constant scalar-drag law is substituted.

2.1.3 Magnetic input and the power balance A transverse field B_y gives the normalized input $u = B_y/B_0$ and interaction potential $-u\sin(d^{\mathsf{T}}q)$. Treating this interaction as external work yields the single physical port

$$\begin{aligned} g(q) &= d\cos(d^{\mathsf{T}}q), \\ y &= g(q)^{\mathsf{T}}v, \\ q' &= \partial_p H_*, \\ p' &= -\partial_q H_* - R_*v + g(q)u. \end{aligned} \tag{6}$$

Consequently $H_*' = -v^{\mathsf{T}}R_*v + uy$ and the SI power is $E_0\omega_0 H_*' = -\dot{q}^{\mathsf{T}}C_{\mathrm{phys}}\dot{q} + \mu B_y\cos(d^{\mathsf{T}}q)d^{\mathsf{T}}\dot{q}$. The distributed-viscosity identification trajectories use $u = 0$; the forced validation and feedback runs apply prescribed or held inputs through this same port. COMSOL Global Equations integrates these four reduced canonical equations in SI time. It does not provide a three-dimensional finite-element rod solution or a PDE-convergence certificate. When a conservative KAM statement is invoked, it concerns the subsystem obtained by setting both the input and dissipation to zero while retaining the same metric and stored potential. The dissipative training reference and the forced control system have their own power and stability statements; conservation is not inferred for either of those flows.

2.2 Full-phase complex energy and dissipative learning

The prescribed kinetic graph is $T_G(q,p) = p^{\mathsf{T}}G(q)^{-1}p/2$, with the positive metric G obtained from the stated finite Gauss reduction. Write

$$\begin{aligned} H_4 &= T_G + V_4(q), \\ \widehat{H}_\vartheta &= H_4 + N_\vartheta(q,p), \\ H_* &= H_4 + U(q). \end{aligned} \tag{7}$$

Here V_4 includes the prescribed quadratic and quartic potential terms. The correction $U(q) = 1 - \cos(s) - s^2/2 + s^4/24$, with $s = d^{\mathsf{T}}q$, defines the conservative reference remainder. It is used in verification; its coefficients are not training labels. The learned correction depends on all four canonical coordinates. Consequently the deployed velocity is

$$\widehat{v} = \partial_p \widehat{H}_\vartheta = G^{-1}p + \partial_p N_\vartheta. \tag{8}$$

2.2.1 The scalar network and its analytic continuation Let $s_j > 0$ be the stored physical input scales, and put $y_j = x_j/s_j$ for $x = (q_1, q_2, p_1, p_2)$. The first layer receives the

four complex features

$$z_j^+ = (y_j + iy_{2+j})/s_0,$$
$$z_j^- = (y_j - iy_{2+j})/s_0, \qquad j = 1, 2,$$

where s_0 is the stored binary64 value of $\sqrt{2}$. On real states the two feature pairs are conjugates. Their coefficients are not tied by conjugacy; their effective values obey the declared raw-parameter constraints. Two width-sixteen complex affine–tanh layers are followed by a bias-free complex linear readout f_ϑ. The model has 736 trainable real coordinates. The real scalar correction is

$$N_\vartheta(x) = \eta \left\{ \frac{\operatorname{Re} f_\vartheta(y) + \operatorname{Re} f_\vartheta(-y)}{2} - \operatorname{Re} f_\vartheta(0) \right\}. \quad (9)$$

For this fit η is the stored energy scale multiplied by 2^{-28}. Evenization and the gauge imply $N_\vartheta(-x) = N_\vartheta(x)$ and $N_\vartheta(0) = 0$. There is no monomial prefactor or separately fitted polynomial replacing the hidden layers.

For complex-domain analysis, $x_1, \ldots, x_4$ are independent complex variables. The displayed linear expressions define $z^\pm$ there; they are not evaluated by conjugating a variable point. If $f_\vartheta^\sharp(y) = \overline{f_\vartheta(\bar{y})}$ denotes coefficient conjugation, replace each real readout in (9) by $(f_\vartheta + f_\vartheta^\sharp)/2$. This gives a holomorphic continuation wherever the actual tanh preactivations avoid their poles. Pole separation is certified for the frozen output on each required domain. The defining layer constraints, input buffers and pre-rounded smooth-absolute constants are reconstructed from the raw checkpoint; rounded exported effective weights do not define a second model.

2.2.2 A separate positive-semidefinite dissipation head A configuration-only complex network with two width-two tanh layers and 42 raw real coordinates produces a lower triangular factor $L_\psi(q)$. With $G = SS^\mathsf{T}$, define

$$\widehat{M}_\psi = \operatorname{Re}(L_\psi L_\psi^*),$$
$$\widehat{R}_\psi = S\widehat{M}_\psi S^\mathsf{T}, \quad (10)$$
$$\widehat{\mathcal{D}}_\psi(q, v) = \tfrac{1}{2} v^\mathsf{T} \widehat{R}_\psi(q) v.$$

For real w, $w^\mathsf{T}\widehat{M}_\psi w = \|L_\psi^* w\|_2^2 \geq 0$, and $\partial_v \widehat{\mathcal{D}}_\psi = \widehat{R}_\psi v$. The deployed canonical decoder is therefore

$$q' = \widehat{v},$$
$$p' = -\partial_q \widehat{H}_\vartheta - \widehat{R}_\psi \widehat{v} + g(q)u, \quad (11)$$
$$g(q) = d \cos(d^\mathsf{T} q).$$

The pair $\widehat{H}_\vartheta + i\widehat{\mathcal{D}}_\psi$ records an energy and a Rayleigh function. The equations are defined by (11), rather than by treating the pair as a single holomorphic Hamiltonian generator. In particular,

$$\widehat{H}'_\vartheta = -\widehat{v}^\mathsf{T} \widehat{R}_\psi \widehat{v} + u\, g^\mathsf{T} \widehat{v}.$$

Both occurrences of $\widehat{v}$ retain $N_{\vartheta,p}$.

2.2.3 The observed weak balance and the executed objective The conservative fit uses fourteen native COMSOL trajectories through time, configuration, observed velocity and prescribed input. The observation chart sets $\Pi = G(q)v$; exported native momenta, momentum derivatives and reference damping values are not optimizer labels. For window w, let Q_w be its declared positive quadrature, h_w its duration, and define

$$r_{4,w} = D_{\dot{p}}^{-1} \frac{\Delta\Pi_w - Q_w[-H_{4,q}(q,\Pi) - \widehat{R}_\psi v + g(q)u]}{h_w},$$
$$r_{\vartheta,w} = r_{4,w} + D_{\dot{p}}^{-1} \frac{Q_w[N_{\vartheta,q} + \widehat{R}_\psi N_{\vartheta,p}]}{h_w}. \quad (12)$$

All network derivatives here are evaluated at (q, Π), and $D_{\dot{p}}$ contains the fixed positive momentum-rate scales. The signs follow by integrating the momentum equation in (11). Centering evaluates the perturbation directly and retains the actual scalar H_4 baseline.

Let n_w be the number of windows and let D_v contain the fixed positive kinematic scales. With the same positive quadrature, the executed full-batch objective is

$$F(\vartheta) = \frac{1}{F_4} \left\{ \frac{1}{2n_w} \sum_w \|r_{\vartheta,w}\|_2^2 + \frac{0.1}{2n_w} \sum_w \frac{Q_w[\|D_v^{-1} N_{\vartheta,p}\|_2^2]}{h_w} \right\}, \quad (13)$$

where $F_4 > 0$ is the fixed initial weak-loss value. The kinematic residual is $N_{\vartheta,p}$ because $H_{4,p}(q,\Pi) = v$ in the prescribed observation chart. No raw-parameter penalty is included in this fit.

The matrix head is already frozen when these conservative coordinates are trained. The two networks are composed sequentially. Full-batch Adam uses 400 updates, learning rate 0.01, and $\epsilon = 10^{-8}$. Every network coordinate remains trainable. The initially zero readout gives the exact initial scalar H_4; hidden-layer changes with a nonzero functional effect are recorded after the second update. Training-only minimization over the initial and recorded objective values selects update 385. The rest-label cohort is used subsequently for reused validation and finite-output certification. It does not enter that checkpoint selection.

The error analysis applies to this stored output and its actual observation residuals. It does not require that Adam has reached a stationary point, nor does it substitute a convergence theorem for the executed optimization. The full-phase real comparison and the earlier factorial study retain their own training budgets and outputs.

2.3 Complex canonical form and symplectic discretization

The complex canonical coordinates are derived from the same full-phase scalar energy as the real equations. Their exact coordinate change is distinct from the stored numerical preprocessing inside the network. The conservative midpoint method is classical (Hairer 2010, Lecture 2, Theorem 3); the formulas below retain both the configuration-dependent inertia and the network's momentum derivatives.

2.3.1 The complex equation of the actual scalar energy

For normalized time τ, set

$$\begin{aligned} z &= (q+ip)/\sqrt{2}, \quad \bar{z} = (q-ip)/\sqrt{2}, \\ q &= (z+\bar{z})/\sqrt{2}, \quad p = (z-\bar{z})/(i\sqrt{2}). \end{aligned} \tag{14}$$

The square root here is the exact mathematical value; the network's fixed input buffer s_0 remains part of its own scalar graph. Put $\mathcal{H}(z,\bar{z}) = \widehat{H}(q(z,\bar{z}), p(z,\bar{z}))$. Treating z and $\bar{z}$ independently in the chain rule gives

$$\partial_z \mathcal{H} = (\widehat{H}_q - i\widehat{H}_p)/\sqrt{2}, \qquad \partial_{\bar{z}} \mathcal{H} = (\widehat{H}_q + i\widehat{H}_p)/\sqrt{2}. \tag{15}$$

Therefore the real canonical equations are equivalent to

$$z' = (\widehat{H}_p - i\widehat{H}_q)/\sqrt{2} = -i\partial_{\bar{z}}\mathcal{H}. \tag{16}$$

The corresponding real symplectic form is $\sum_j dq_j \wedge dp_j = i\sum_j dz_j \wedge d\bar{z}_j$.

Write $W(q) = G(q)^{-1}$, $v_* = W(q)p$ and $G_j = \partial_j G$. Differentiating $WG = I$ gives $\partial_j W = -WG_jW$. For the deployed scalar $\widehat{H} = p^{\mathsf{T}}Wp/2 + V_4 + N(q,p)$, its complete derivatives are

$$\widehat{H}_p = v_* + N_p, \qquad \widehat{H}_{q_j} = -\tfrac{1}{2}v_*^{\mathsf{T}}G_j v_* + \partial_j V_4 + N_{q_j}. \tag{17}$$

Thus the configuration derivative of inertia and the full neural gradient enter (16). In particular, v_* is the reference velocity; it is not the full learned velocity.

The physical real scalar generally depends on both z and $\bar{z}$. Its analytic continuation instead extends q, p to independent complex coordinates Q, P. The two linear combinations $(Q+iP)/\sqrt{2}$ and $(Q-iP)/\sqrt{2}$ are then independent coordinates. For the actual network, the continuation $N^{\mathbb{C}}$ uses coefficient-sharp readouts as defined in Section 2.2, giving

$$\widehat{H}^{\mathbb{C}}(Q,P) = \tfrac{1}{2}P^{\mathsf{T}}G(Q)^{-1}P + V_4(Q) + N^{\mathbb{C}}(Q,P). \tag{18}$$

The domain must exclude singularities of G^{-1} and the literal tanh activations. Matrix transposes in this continuation are ordinary analytic transposes. Taking the real part of a variable complex evaluation would not define the required holomorphic continuation.

The configuration-only Rayleigh head satisfies $\partial_v \widehat{\mathcal{D}} = \widehat{R}(q)v$. Its canonical decoder evaluates this derivative at $v = \widehat{H}_p$. Adding the magnetic port therefore yields

$$z' = -i\partial_{\bar{z}}\mathcal{H} + \frac{i}{\sqrt{2}}\{-\widehat{R}(q)\widehat{H}_p + ug(q)\}. \tag{19}$$

The time discretization below concerns the conservative subsystem obtained by setting the Rayleigh force and input to zero.

2.3.2 Implicit midpoint with all momentum derivatives retained

Let $x = (q,p)$, $J_c = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$ and $f = J_c\nabla\widehat{H}$. A midpoint evaluation defines

$$x_{n+1} = x_n + hJ_c\nabla\widehat{H}(m_n), \qquad m_n = (x_n + x_{n+1})/2. \tag{20}$$

Writing $\bar{q} = (q_n + q_{n+1})/2$, $\bar{p} = (p_n + p_{n+1})/2$ and $\bar{v}_* = G(\bar{q})^{-1}\bar{p}$ gives the actual coupled equations

$$\begin{aligned} q_{n+1} &= q_n + h\{\bar{v}_* + N_p(\bar{q},\bar{p})\}, \\ p_{n+1,j} &= p_{n,j} + h\{\tfrac{1}{2}\bar{v}_*^{\mathsf{T}}G_j(\bar{q})\bar{v}_* - \partial_j V_4(\bar{q}) - N_{q_j}(\bar{q},\bar{p})\}. \end{aligned} \tag{21}$$

Equivalently, (14) gives the complex midpoint step

$$z_{n+1} = z_n - ih\partial_{\bar{z}}\mathcal{H}((z_n + z_{n+1})/2, (\bar{z}_n + \bar{z}_{n+1})/2).$$

For a closed ball $\overline{B}(x,b)$ in a real domain with $\widehat{H} \in C^3$, suppose $\|f\| \le M$ and $\|Df\| \le L$. If

$$|h|M < b, \qquad |h|L/2 < 1, \tag{22}$$

then $y \mapsto x + hf((x+y)/2)$ maps that ball into itself and is a contraction. Its unique fixed point defines a differentiable local map Φ_h by the implicit-function theorem.

Proposition 1. Symplecticity of the conservative midpoint map. *Under (22), the exactly solved local step satisfies* $(D\Phi_h)^{\mathsf{T}}J_cD\Phi_h = J_c$.

Proof. At its midpoint put $A = J_c \nabla^2 \widehat{H}(m_n)$. Symmetry of the full scalar Hessian gives $A^{\mathsf{T}} J_c + J_c A = 0$. Differentiation of (20) yields

$$(I - hA/2) D\Phi_h = I + hA/2.$$

The matrices $U_h = I - hA/2$ and $V_h = I + hA/2$ commute, and

$$V_h^{\mathsf{T}} J_c V_h = J_c + (h^2/4) A^{\mathsf{T}} J_c A = U_h^{\mathsf{T}} J_c U_h.$$

Since $D\Phi_h = V_h U_h^{-1}$, multiplication on the left by $U_h^{-\mathsf{T}}$ and on the right by U_h^{-1} proves the claim. No separability or vanishing momentum derivative of N was used.

2.3.3 Numerical error and its distinction from learning error Use the maximum-component norm in physical canonical coordinates, its induced matrix norm and the corresponding bilinear norm. Let a common convex neighborhood contain all exact and numerical states and segments under comparison, and let M, L, B_2 bound $f, Df, D^2 f$ in these norms. The observational field bound δ_0 below uses this same physical-coordinate maximum norm. Choose $h_0 > 0$ with $h_0 L/2 = \kappa < 1$, require $0 < h \le h_0$, and put

$$C_d = \frac{B_2 M^2}{24} + \frac{L^2 M}{6}, \qquad C_\ell = \frac{C_d}{1-\kappa}, \qquad \Lambda = \frac{L}{1-\kappa}.$$

The exact endpoint's midpoint-equation defect is at most $C_d h^3$. Contraction then bounds the one-step state error by $C_\ell h^3$. The step's Lipschitz factor is at most $(1 + hL/2)/(1 - hL/2) \le e^{\Lambda h}$, so for equal initial states and $nh \le T$,

$$\|\Phi_h^n(x_0) - \phi_{\widehat{H}}^{nh}(x_0)\| \le C_\ell h^2 \frac{e^{\Lambda T} - 1}{\Lambda}. \tag{23}$$

The quotient is T when $\Lambda = 0$. Supplementary Section S1 proves these constants by integral remainders and includes finite nonlinear-solver residuals. Such residuals need not preserve the exact symplectic identity.

By comparison, if $\|X_{\widehat{H}} - X_{H_*}\| \le \delta_0$ and X_{H_*} has Lipschitz bound L_* on a common trajectory neighborhood, the equal-initial-state conservative flows satisfy

$$\|\phi_{\widehat{H}}^t(x_0) - \phi_{H_*}^t(x_0)\| \le \delta_0 \frac{e^{L_* t} - 1}{L_*}. \tag{24}$$

This follows from the integral difference equation and Gronwall's inequality, with quotient t when $L_* = 0$. Its error term is the actual learned-field discrepancy, which is bounded from observations below. The numerical and learned-model errors therefore enter separate terms; a small time step cannot remove an inaccurate learned field.

For a quadratic energy the exact midpoint rule preserves that energy, but the deployed nonquadratic scalar generally has an additional energy increment. The full learned dissipative vector field has canonical divergence

$$-\operatorname{tr}(\widehat{R}\widehat{H}_{pp}) + g(q)^{\mathsf{T}} \nabla_p u.$$

For prescribed input the second term vanishes. The first uses the actual Hessian $\widehat{H}_{pp} = G^{-1} + N_{pp}$, rather than replacing it by the reference inverse metric. For prescribed input, a strictly positive trace implies volume contraction and excludes symplecticity of that dissipative flow. With state-dependent feedback, the corresponding sufficient condition is $\operatorname{tr}(\widehat{R}\widehat{H}_{pp}) - g(q)^{\mathsf{T}} \nabla_p u > 0$. The certified KAM conclusions concern the continuous conservative flow; a discrete-map KAM statement would require its own step-size nonresonance, map geometry and invariance-defect certificate.

2.3.4 Computable discrete forms with the learned damping For a prescribed held command, write the complete learned field as $f = (\widehat{H}_p, -\widehat{H}_q + F)$, where $F = -\widehat{R}\widehat{H}_p + ug$. Its implicit midpoint map Φ_h is a local diffeomorphism under a self-map condition and $|h| \sup \|Df\|/2 < 1$. With $A = Df$ at the midpoint and $B = I - hA/2$, its exact derivative satisfies

$$D\Phi_h^{\mathsf{T}} J D\Phi_h - J = hB^{-\mathsf{T}} \begin{pmatrix} F_q - F_q^{\mathsf{T}} & F_p \\ -F_p^{\mathsf{T}} & 0 \end{pmatrix} B^{-1}.$$

Thus the actual damping contributes a canonical two-form defect. A time-indexed form can instead be computed by $\Omega_{n+1} = (\Phi_n^{-1})^* \Omega_n$, using inverse midpoint and its derivative, so that $\Phi_n^* \Omega_{n+1} = \Omega_n$. Closedness and nondegeneracy follow from the pullback of the full form field. This is an application of standard differential-form transport, with all learned momentum and damping-matrix derivatives retained. Supplementary Section S24 proves the local construction and its consistency with continuous form transport. A ten-step numerical prototype for the frozen GC–PSD composition checks the derivative and pullback identities; it is separate from the validated physical-control integration.

3 Numerical approximation and observation-based learning error

3.1 Archived comparison protocols and the certified output

The four-cell factorial and the later precision-centered comparison use related architectures but different objectives, output scales and checkpoints. The labels R, C, GR and GC in the factorial tables refer only to that protocol. The deployed GC in (7) is the selected output of the precision-centered comparison. We denote that certified output by $\widehat{H}_0$ at θ_0. Function-specific error, KAM and feedback statements refer to this checkpoint, while the additional outputs $\widehat{H}_R^J$ and $\widehat{H}_C^J$ in Section 3.5 have separate parameter identities and do not inherit those certificates.

3.2 Generic phase-space networks: representation and quadratic prior

A factorial experiment compares real and complex networks with and without a fixed quadratic Hamiltonian. It uses the current finite-Gauss mechanics and distributed-viscosity observations. The four cells are R (real, no fixed quadratic term), C (complex, no fixed quadratic term), GR (real with the quadratic prior), and GC (complex with the quadratic prior). The quadratic prior improves both paired representations, and GC has the smallest factorial error for every seed. The absolute errors nevertheless remain substantially larger than those of the shared fourth-order model with the network correction set to zero.

Writing $G_0 = G(0)$ and $\mathsf{K}_2 = B + dd^T$, every cell has the form

$$\begin{aligned} H_s(q,p) &= T_{\mathrm{nl}}(q,p) + V_{\mathrm{quartic}}(q) \\ &\quad + I_s H_2(q,p) + N_{\theta,s}(q,p), \\ T_{\mathrm{nl}}(q,p) &= \tfrac{1}{2} p^T [G(q)^{-1} - G_0^{-1}] p, \\ V_{\mathrm{quartic}}(q) &= -\tfrac{1}{24} (d^T q)^4, \\ H_2(q,p) &= \tfrac{1}{2} p^T G_0^{-1} p + \tfrac{1}{2} q^T \mathsf{K}_2 q, \end{aligned} \tag{25}$$

where $I_s = 0$ for R and C and $I_s = 1$ for GR and GC. Thus the ablation removes the complete origin quadratic Hamiltonian, including its momentum block. The known nonlinear kinetic remainder and quartic magnetic term are common to all cells; R and C are not models without any prescribed mechanics.

The generic residual acts on all four canonical coordinates $x = (q,p)$. Its inputs use componentwise RMS scales computed from the fourteen training trajectories, and its energy scale is the training RMS of H_2. A common evenization and subtraction at the origin impose $N_{\theta,s}(-x) = N_{\theta,s}(x)$ and $N_{\theta,s}(0) = 0$. No degree-six prefactor or independent quadratic skip is present. In particular, the residual Hessian is free, so the no-prior cells are not structurally prevented from approximating the missing quadratic term. This is an approximation task for finite-width networks, not a claim of exact quadratic representation at arbitrary parameter values.

The real network has two width-24 hidden layers with real tanh and a linear readout, giving 744 trainable real coordinates. The complex network has two width-16 hidden layers with literal complex tanh, independent $z, \bar{z}$ affine inputs, and a real readout, giving 736 real coordinates. Its layerwise imaginary-argument budgets are 0.15 and 0.6. These constraints apply on the declared normalized phase box containing all training states. The counts differ by approximately 1.09%; there is no parameter padding. Widths, parameter domains and initialization laws also differ. Within a representation and seed, the prior and no-prior cells start with identical raw network parameters; real and complex initial functions are not matched.

All cells use the same frozen 42-coordinate complex PSD matrix head and the same distributed-viscosity cohort of fourteen trajectories on $[0, 0.4]$, with 801 observation nodes per trajectory. The earlier configuration-only structured model used a different conservative $[0, 0.2]$ cohort; it is a separate baseline. The matrix head is the previously identified distributed-viscosity head, fitted with that structured Hamiltonian frozen; it is not refitted for these comparators. Only time, configuration, observed velocity and prescribed input enter the fit. Observed momentum is reconstructed as $G(q)v$. The predicted dynamics use

$$\begin{aligned} q' &= H_{s,p}, \\ p' &= -H_{s,q} - \widehat{R}(q) H_{s,p} + u g(q). \end{aligned} \tag{26}$$

Consequently, dissipation uses the model's actual $H_{s,p}$, and the kinematic discrepancy is $H_{s,p} - v$; observed velocity is not silently substituted into the predicted damping force. Twenty windows per trajectory use the same actual-time, five-point Newton–Cotes quadrature. The common scalar loss is the mean normalized momentum-weak square plus 0.1 times the quadrature-weighted normalized kinematic square and a raw-coordinate penalty with coefficient $10^{-20}/2$. All scales come from the fourteen training trajectories. No separate potential-nuisance fit, reference-force labels or validation selection is used.

Seeds 43, 47 and 53 give twelve fits. Each completes 400 full-batch Adam updates with learning rate 0.01, $(\beta_1, \beta_2) =$

$(0.9, 0.999)$ and $\epsilon = 10^{-8}$. The lowest finite training total among the initial state and the 400 recorded updates selects each checkpoint. All twelve selected outputs are frozen before this factorial evaluation, and their training-loss replays agree. There are no failed fits, no failed forty-point model evaluations and no failed model–holdout evaluations among the 48 latter combinations. The three seeds are optimization repeats on common data, rather than independent physical datasets. The initial functions also differ materially between representations: for seed 43, the initial training totals for R, C, GR and GC are 1.197470, 1.085593, 0.139463 and 0.001348, respectively. The common learning rate and update budget therefore do not remove the effects of the different initialization and constraint-induced function scales, especially in the prior-assisted pair.

The primary evaluation uses the same forty sealed observation-only gradient labels Y_i at the prescribed rest points $(Q_i, 0)$:

$$E_s^2 = \frac{1}{40} \sum_{i=1}^{40} \left\| H_{s,q}(Q_i, 0) - Y_i \right\|_2^2. \tag{27}$$

The forty labels and four original holdout trajectories had already served other analyses. They are therefore reused validation, not a new blind confirmation of these generic comparators. At $p = 0$, the metric tests potential gradients; it does not validate arbitrary phase-space derivatives. Table 1 retains every seed.

The prior improves both representations in every paired seed, and the complex fit has lower RMSE within both prior settings. The interaction is negative for every seed: the prior produces the larger absolute error reduction for the real network and narrows the absolute real–complex gap. These observations describe this data, architecture pair and optimization budget. They do not isolate complex arithmetic from the different widths, constraints, initialization and parameter-dependent regularization, or establish universal complex superiority.

Full-phase consistency is assessed separately on the four reused holdout trajectories, using the fixed training scales, the same frozen matrix head and no additional coefficients. Table 3 pools squared normalized errors equally across the three seeds and four trajectories before taking the square root. It reports weak momentum and quadrature-weighted $H_{s,p} - v$ discrepancies, not a new integrated rollout. The last column gives the range across seeds of $\{40^{-1} \sum_i \|H_{s,p}(Q_i, 0)\|_2^2\}^{1/2}$. Although the reference velocity vanishes at these rest points, the generic residual can contain mixed q, p terms. Central evenization does not impose evenness in p at fixed q, and the nonzero rest velocity diagnostic remains relevant even when the potential-gradient error is relatively small.

The absolute-accuracy limitation is substantial. On the identical forty labels, the zero-network fourth-order baseline $H_4 = T_{\mathrm{nl}} + H_2 + V_{\mathrm{quartic}}$ has gradient-vector RMSE approximately 1.297917×10^{-11}. Every generic GC fit is worse: its RMSE ranges from 7.966939×10^{-6} to 1.098578×10^{-5}, approximately 6.1×10^5 to 8.5×10^5 times the baseline error. The generic prior-assisted family contains H_4 at zero residual, so this gap is a limitation of the stated generic training protocol at the resolved signal scale, not proof that the physical prior is unrepresentable. Relative improvement within the factorial experiment therefore does not demonstrate improvement over the prescribed fourth-order physics. The precision-centered comparison below tests whether output scaling and centering reduce this absolute-accuracy gap with the full-phase network.

These comparisons use ordinary floating-point network evaluation. Exact rational propagation of the sealed label widths supplies sensitivity bounds conditional on the returned network numbers; it does not enclose neural evaluation and differentiation errors and is not a strict network-ranking certificate. These twelve factorial outputs are distinct from the selected precision-centered GC output certified in Section 4. Their relative ranking does not transfer any function-specific certificate to them.

3.3 A precision-centered prior-assisted comparison

A separate two-fit comparison tests whether objective and output scaling account for part of the absolute-accuracy gap in Section 3.2. It retains the generic full-phase real and complex architectures, with 744 and 736 trainable real coordinates, respectively, and uses seed 43 only. Both models now have the actual scalar form $H_4 + N_s$, with the same finite-Gauss kinetic energy and fourth-order potential. An exactly zero readout makes their initial functions identical to H_4; there is no degree-six multiplier, polynomial replacement or separately fitted potential block.

The output multipliers are $\rho_{\mathrm{GR}} = 2^{-32}$ and $\rho_{\mathrm{GC}} = 2^{-28}$. They are dyadic scales derived from the training residual and readout-Jacobian singular values, using only the same fourteen distributed-viscosity trajectories. The momentum-weak balance is centered about the actual scalar H_4 before adding the network perturbation. The perturbation is evaluated directly, rather than obtained by subtracting full

Table 1. Generic-network gradient-vector RMSE on the same forty reused rest-point labels. Entries are $10^3 E_s$, in common dimensionless potential-gradient units. The final row is the arithmetic seed mean.

Seed	R	C	GR	GC
43	7.219805	6.034963	0.099531	0.009317
47	9.520357	6.648840	0.298038	0.007967
53	12.884718	5.769706	1.104723	0.010986
Mean	9.874960	6.151169	0.500764	0.009423

Table 2. Five paired contrasts of the RMSE in Table 1, multiplied by 10^3. For $A - B$, a positive value means that B has smaller error. The interaction is $(\mathrm{GR} - \mathrm{GC}) - (\mathrm{R} - \mathrm{C})$. Means use all three paired seeds.

Contrast	Seed 43	Seed 47	Seed 53	Mean
R−GR	7.120274	9.222319	11.779995	9.374196
C−GC	6.025646	6.640873	5.758720	6.141746
R−C	1.184843	2.871517	7.115013	3.723791
GR−GC	0.090214	0.290071	1.093737	0.491341
Interaction	−1.094628	−2.581446	−6.021276	−3.232450

Table 3. Reused full-phase diagnostics. Weak and kinematic columns are pooled normalized RMS values over twelve seed–trajectory combinations per cell. The rest-velocity range covers the three seedwise forty-point RMS values and is in the original dimensionless velocity coordinates.

Cell	Momentum weak	$H_{s,p} - v$	Rest $H_{s,p}(Q,0)$ RMS range
R	5.310319×10^{-2}	1.255422×10^{-1}	$[1.518317 \times 10^{-3},\ 5.222680 \times 10^{-3}]$
C	3.948955×10^{-2}	6.424029×10^{-2}	$[1.484900 \times 10^{-3},\ 2.417393 \times 10^{-3}]$
GR	4.187033×10^{-3}	1.810586×10^{-2}	$[1.498658 \times 10^{-4},\ 3.086894 \times 10^{-3}]$
GC	5.695685×10^{-5}	3.301787×10^{-4}	$[8.301210 \times 10^{-6},\ 2.276685 \times 10^{-5}]$

and baseline floating-point forces. The baseline still contains the arithmetic of the native observations and quadrature. Its quadratic jet is evaluated from the actual scalar H_4, rather than regrouped through the stored approximation to K_2. The sum of the normalized weak loss and the 0.1-weighted kinematic loss is divided by its fixed H_4 value. There is no raw-parameter penalty. The matrix head remains fixed, and the kinematic and damping terms retain the actual full-phase $H_{s,p}$.

Both fits complete 400 full-batch Adam updates with learning rate 0.01 and $\epsilon = 10^{-8}$. Every layer remains trainable. The initial zero readout gives zero hidden-layer data gradients at the first update; nonzero hidden-parameter and prediction changes are recorded after the second update. Checkpoint selection again minimizes the training total over the initial state and all recorded updates. The selected updates are 400 for GR and 385 for GC, with training totals relative to the initial value of approximately 4.1951×10^{-4} and 1.9017×10^{-2}, respectively.

Both selected outputs are frozen before their evaluation against the same forty previously used labels. No additional held-out dataset is introduced or used for selection in this comparison. It is exploratory reused validation, rather than a new blind test, and does not replace the twelve fits or five paired contrasts above. Only these forty labels are evaluated; no trajectory-holdout evaluation is performed for the two repaired outputs.

The real model has the smaller error in this comparison, and neither learned output improves on the shared prior (Table 4). Together with the original factorial result, this reversal demonstrates sensitivity to the stated training scales and initialization; it does not establish an intrinsic advantage of either representation. The selected precision-centered GC output is the frozen function in (7). Its subsequent observation-error, conservative-torus and physical-feedback certificates are proved for this checkpoint in Sections 3.6, 4 and 5. They do not use a certificate belonging to the earlier structured model.

3.4 A paired full-phase check against the physical prior

The original certified GC output $\widehat{H}_0$ is compared with H_4 on the moving anchors of Section 3.9. This is a postfit diagnostic on the existing independent-anchor record; the comparison and metric were not specified before those anchors were observed. This archived comparison refits neither model. The reference is the same finite-Gauss, full-cosine energy $H_* = H_4 + U(q)$ used in the conservative certificate.

Because U is independent of momentum, its two configuration derivatives at the 49 distinct configuration pairs supply the reference correction at every moving anchor. The already sealed neural gradients are evaluated at their actual four-dimensional coordinates. The conservative field errors

Table 4. Precision-centered comparison on the same forty reused rest labels. Gradient-vector RMSE uses the normalization in (27). Neural derivatives are ordinary floating-point scalar-energy automatic derivatives; the table is not an interval ranking certificate.

Model	Selected update	Rest-gradient vector RMSE
GR	400	$1.427681587 \times 10^{-11}$
GC	385	$9.819100531 \times 10^{-11}$
Shared H_4	—	$1.297916885 \times 10^{-11}$

are

$$X_{\mathrm{GC}} - X_* = J(\nabla N - (U_q, 0)),$$
$$X_4 - X_* = J(-U_q, 0).$$

The canonical matrix J is Euclidean orthogonal. The squared error is therefore the sum of the squared force and velocity errors in the original dimensionless canonical coordinates, without the coordinate rescalings used in the observation inverse.

Let m_i be the observed multiplicity of a moving support point and π_i its sampling density relative to the finite uniform law. We report the importance-weighted vector error

$$E_X^2 = \frac{1}{15368} \sum_i \frac{m_i}{\pi_i} \|X_{\mathrm{model}}(x_i) - X_*(x_i)\|_2^2. \qquad (28)$$

All 15,368 moving draws and their 2,394 distinct support points are retained. The weights sum to approximately 0.9966912; they are not renormalized to one. This realized estimate is not the exact mean on the 2,401-point population: seven support points were not evaluated.

The GC-to-H_4 ratio of full-field RMS errors lies strictly between 22.34148 and 22.34150. Its velocity channel accounts for approximately 14.459% of the GC squared field error; this reference channel is exactly zero for the correction to H_4. The largest unweighted vector errors over the 2,394 evaluated nodes are approximately 1.64950×10^{-11} for H_4 and 1.34629×10^{-10} for GC. Thus the rest-gradient accuracy limitation also appears in this full-phase comparison. These paired results concern the frozen GC output and the stated numerical reference. They concern the original checkpoint and do not establish a ranking over all continuous states. Section 3.5 separately reports the subsequently fitted real and complex outputs.

The original native training nodes give a different comparison. Using positive quadrature-time weights, the full-field vector RMS errors are 1.121863×10^{-10} for H_4 and 4.460762×10^{-11} for GC, a ratio of approximately 0.397621. These are training-node diagnostics, with a different state distribution from the finite anchor law. The integrated residual decomposition in Supplementary Section S14 separates the reference conservative correction, damping mismatch and discrete reference defect. Their coordinate RMS magnitudes in raw window momentum-rate units are approximately 7.92245×10^{-11}, 7.79015×10^{-12} and 4.22328×10^{-15}, respectively. The learned integrated correction has cosine similarity approximately 0.960116 with the true conservative correction. Thus this record resolves a conservative learning signal, while the two comparisons expose a limitation in its transfer across the tested state distributions.

The low-order jet also differs from that of the prescribed correction. A directed evaluation gives $\partial_{11} N(0) > 5 \times 10^{-9}$, whereas $D^2 U(0) = 0$. Supplementary Section S15 proves that, along a specified small configuration axis, the learned field error has a nonzero term of order t while the H_4 error is of order t^5. This identifies a local loss of the prior's approximation order. It does not by itself attribute the full moving-anchor comparison to a single mechanism.

3.5 A physical low-order penalty and a matched-scale comparison

The local order obstruction above motivates a further identification experiment. The prescribed reference correction satisfies

$$U(q) = 1 - \cos(d^T q) - \tfrac{1}{2}(d^T q)^2 + \tfrac{1}{24}(d^T q)^4,$$

so its second- and fourth-order jets vanish. We use this known mechanical property as a soft prior, retaining the ordinary full-phase real and complex networks. Denote the new outputs by $\widehat{H}_R^J = H_4 + N_R^J$ and $\widehat{H}_C^J = H_4 + N_C^J$. They are distinct from the certified checkpoint $\widehat{H}_0 = H_4 + N_0$ at parameter θ_0. In the observation, KAM and feedback applications below, $\widehat{H}$ and N without the superscript J continue to denote this original certified function.

Let $y_j = x_j / s_j$, using the common training-state scales, and let E be the actual shared energy scale. The ordinary Taylor coefficients of $N^J(Sy)/E$ of total degrees two and four form vectors c_2 and c_4 of lengths 10 and 35. These are coefficients with factorials divided out. On the fixed known rectangle $y \in \prod_j [-b_j, b_j]$, define

$$(A_m)_{\alpha\gamma} = \frac{E^2}{4F_4} \sum_{j=1}^{4} \frac{\omega_j}{s_j^2 a_j^2} \mathbb{E}[\partial_j y^\alpha \partial_j y^\gamma], \qquad |\alpha| = |\gamma| = m, \qquad (29)$$

Table 5. Postfit paired comparison on the existing moving-anchor record. Values are importance-weighted vector RMS errors. Directed interval propagation of the sealed labels and gradients establishes the reported ranking for this record.

Model	Force error	Velocity error	Full field error
Shared H_4	4.052624×10^{-12}	0	4.052624×10^{-12}
Selected GC	8.374046×10^{-11}	3.442855×10^{-11}	9.054165×10^{-11}

where $F_4 > 0$ is the original H_4 native-loss baseline, $\omega = (1, 1, \kappa, \kappa)$, and κ is the stored value of 0.1. The first two a_j are the original momentum-rate scales and the last two are the original velocity scales. Thus the Gram penalty measures low-order conservative force and velocity in the original observation units. Its exact entries use uniform-box monomial moments. The matrices are positive definite because a nonconstant homogeneous polynomial cannot have zero gradient on an open box. The implemented symmetric binary64 matrices $\widetilde{A}_m$ are separately verified as positive definite exact dyadic matrices, with relative Loewner rounding bounds below 8.277×10^{-17} and 1.200×10^{-15} for degrees two and four.

The fixed training objective is

$$\begin{aligned} \mathcal{J} &= F_{\mathrm{nat}}/F_4 + \mathcal{R}_2 + \mathcal{R}_4, \\ \mathcal{R}_m &= c_m^T \widetilde{A}_m c_m, \qquad m = 2, 4. \end{aligned} \tag{30}$$

Here F_{nat} is the centered native weak-momentum and kinematic loss, retaining $N_q + \widehat{R}N_p$ and N_p respectively. Both penalty weights are one, fixed without oracle-label tuning. The coefficients are differentiated through the original constrained affine maps. Supplementary Sections S17 and S18 give the finite coefficient algebra and the physical Gram construction. A bound on this objective controls the returned output's low-order polynomial derivatives; it does not control the higher-order remainder or impose exact zero jets. Supplementary Section S19 constructs a nonzero exact-zero-low-jet member of the original complex class, without claiming that training reaches that member or fits the reference.

One fit per architecture uses seed 43, exact zero readouts at the common initial function H_4, and the same output scale $E = 4.769815841012403 \times 10^{-12}$. The original 744 real and 736 complex real-coordinate parameters remain active. Each fit receives 400 full-batch projected-Adam updates with learning rate 0.01, $\epsilon = 10^{-8}$, and $(\beta_1, \beta_2) = (0.9, 0.999)$. After each update all raw coordinates are clipped to $[-2, 2]$; the optimizer moments are retained. The metric, damping head and fourteen training trajectories remain fixed. Only their original (τ, q, v, u) observations enter fitting. Selection uses the minimum finite complete training objective, giving updates 397 and 400 for the real and complex outputs. The oracle anchors enter neither fitting nor checkpoint selection.

For the native record the reported vector RMS is

$$E_{\mathrm{nat}}^2 = \frac{\sum_{i=1}^{11480} w_i \|X_{\mathrm{model}}(x_i) - X_*(x_i)\|_2^2}{\sum_i w_i}, \tag{31}$$

where $w_i > 0$ are the original actual-time quadrature weights divided by their window durations. Repeated window endpoints retain their weights. The same stable full-sine reference is used for all four rows; the earlier reference array differs by at most 4.136×10^{-25} from the new stored binary64 reference centers. An equal- appearance average is a separate diagnostic, and is not mixed with (31). The vector field is formed using the full N_p, and centered error differences avoid subtracting large mechanical forces.

At every moving anchor, each new output's four neural derivatives are recomputed from its own frozen raw weights at 512-bit precision. The reference configuration labels and the original multiplicities and importance weights are reused; original neural gradients are not substituted for the new functions. Table 6 shows that both new outputs improve on H_4 on these two records. For the moving anchors, $\widehat{H}_C^J$ has approximately 53.2% smaller full-field RMS error than H_4 and 5.0% smaller error than $\widehat{H}_R^J$. The paired squared-error difference has a strict sign under directed propagation. However, its anchor velocity error is larger than the real output's error. On the native record the complex velocity error is instead smaller. The comparison therefore does not establish superiority in every channel.

These are exploratory comparisons after the anchor record and the original output's limitations were known. The comparison with $\widehat{H}_0$ changes regularization, output scaling and the optimization protocol together. It cannot isolate one change as the cause of improvement. The two new architectures share a scale and update budget but differ in width, topology and parameter count; one optimization seed is insufficient for a general representation claim. No full-network population-risk, conservative-torus or physical-control certificate is inferred for either new output from these finite diagnostics. The existing certificates remain attached to θ_0.

The new complex output separately passes an actual-parameter pole test on $|\Re x_j| \le .06$, $|\Im x_j| \le .01$ and the collar $(.061, .011)$; its second-layer guard bounds are below 1.014874 and 1.096925, respectively, both below $\pi/2$.

Table 6. Full-phase reference errors for the shared prior, original certified complex output, and two additional jet-regularized outputs. All entries are vector RMS errors in units of 10^{-12} in the same canonical coordinates. Native errors use the 11,480 original window appearances with positive quadrature-time weights normalized by their stored sum. Anchor errors use (28), with no self normalization. Values are rounded; the two records have different state distributions and are compared within each record.

	Native	Reused moving anchors		
Model	Full field	Force	Velocity	Full field
H_4	112.1863	4.052624	0	4.052624
$\widehat{H}_0$	44.60762	83.74046	34.42855	90.54165
$\widehat{H}_R^J$	9.842122	1.974094	0.2996202	1.996703
$\widehat{H}_C^J$	9.051897	1.788505	0.6336800	1.897446

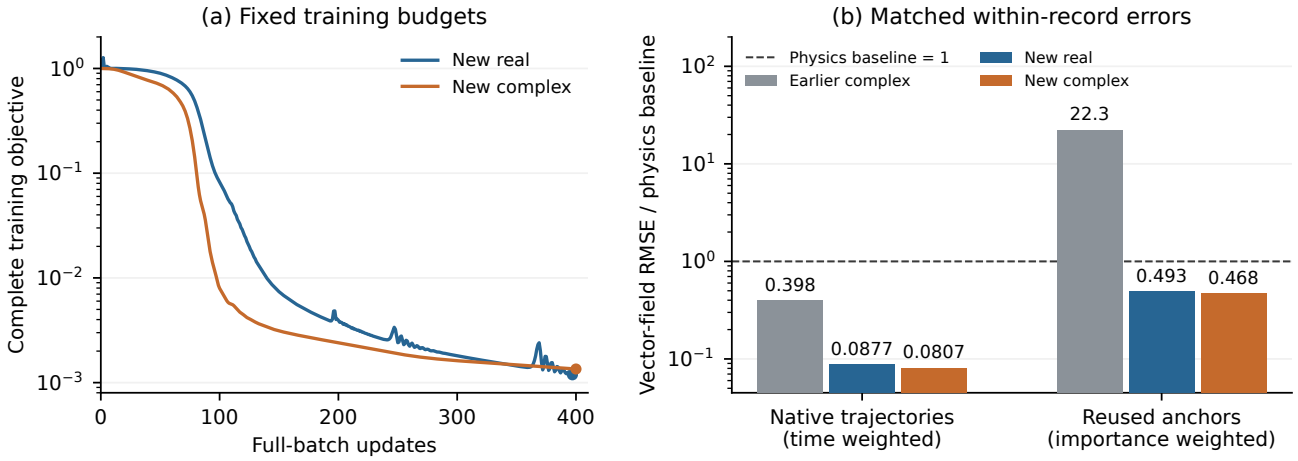


Figure 1. The fixed two-fit experiment and its reference comparisons. Left: the complete native-plus-jet training objectives and selected checkpoints. Right: vector RMS errors relative to H_4, separately for the positive-time-weighted native record and the importance-weighted moving-anchor record. The original certified complex output is retained in the comparison. The records have different state distributions; ratios are interpreted within each record. The reused postfit evaluations are not fresh test data, confidence intervals, or evidence of a causal advantage of complex arithmetic.

Supplementary Section S20 also bounds the derivatives of N_C^J through order three. These are bounds for the correction, not for $N_C^J - U$. The real output's coarse second-layer guard is unresolved; this is not a verified pole or a failure of actual holomorphy.

3.6 A deterministic observation-to-function error bound

An empirical weak loss controls observations along the available curves. To obtain an error on a complex state domain, we additionally certify an observation inverse and the actual remainder of the selected network. Throughout this subsection the output is the same 736-coordinate complex model used in the deployed energy. Set

$$e = N_\vartheta - U,$$
$$\Omega = \left\{ x \in \mathbb{C}^4 : \begin{array}{l} |\operatorname{Re} x_j| \le 3/50, \\ |\operatorname{Im} x_j| \le 1/100 \end{array} \right\}.$$

The functions have analytic extensions to a neighbourhood of this box.

Let χ contain every nonconstant monomial of even total degree at most six in the normalized coordinates x_j/s_j. There are 129 columns: 15 configuration-only columns and 114 containing a momentum factor. Choose an explicitly specified polynomial $P_N = \chi^{\mathsf{T}} a_N$ and write

$$\begin{aligned} U &= U_6 + \rho_U, \\ P_N &= N_\vartheta - \rho_N, \\ U_6 &= \frac{(d^{\mathsf{T}} q)^6}{720}. \end{aligned} \tag{32}$$

The polynomial is an auxiliary representative of the fitted function; the trained network and its hidden layers remain unchanged.

The observation operator $\mathcal{L}$ uses configuration derivatives at rest anchors $(Q_i, 0)$ and momentum derivatives at the represented chart points $(\bar{q}_l, G(\bar{q}_l)\bar{v}_l)$. At rest, sealed labels satisfy $Y_i = \nabla(V_4 + U)(Q_i) + \xi_i$. The signed residuals are

$$\begin{aligned} r_{q,i} &= N_{\vartheta,q}(Q_i, 0) \\ &\quad + V_{4,q}(Q_i) - Y_i, \\ r_{p,l} &= N_{\vartheta,p}(\bar{q}_l, G(\bar{q}_l)\bar{v}_l). \end{aligned} \tag{33}$$

Only the selected components are retained. Since $U_p = 0$, the second reference label is structural information from the shared kinetic graph; it does not require exported native momentum or assert that the chart point equals an exact physical trajectory state. With zero moving entries in ξ, the exact identity is $r = \mathcal{L}e - \xi$.

The selected physical derivative rows form a square design $A = \mathcal{L}\chi^{\mathsf{T}}$. After a common column permutation, $A = A_q \oplus A_p$ with sizes 15 and 114. Configuration derivatives at $p = 0$ annihilate mixed monomials, while momentum derivatives annihilate configuration-only monomials. Interval evaluation encloses the exact observed chart, and a fixed exact dyadic matrix R satisfies

$$\begin{aligned} \sup_A \|I - RA\|_\infty \le \eta_A < 1, \\ \mathcal{J}_A = (RA)^{-1}R, \\ \kappa_A = \frac{\eta_A}{1-\eta_A}. \end{aligned} \tag{34}$$

In particular, $\mathcal{J}_A A = I$. Physical derivative rows retain their $1/s_j$ factors. If row weights are introduced, the same weights must multiply residuals, label errors and point remainders.

Put $t_N = \mathcal{L}\rho_N$, $t_U = \mathcal{L}\rho_U$ and $F_\alpha = \partial_x^\alpha \chi^{\mathsf{T}}$. Applying $\mathcal{L}$ to (32) and using the residual sign gives

$$\begin{aligned} A(a_N - a_U) &= r + \xi - t_N + t_U, \\ \partial_x^\alpha e &= F_\alpha \mathcal{J}_A (r - t_N + t_U) \\ &\quad + F_\alpha \mathcal{J}_A \xi \\ &\quad + \partial_x^\alpha \rho_N - \partial_x^\alpha \rho_U. \end{aligned} \tag{35}$$

Here a_U is the coefficient vector of U_6. It is used only in verification. Both observation-point remainders and target-domain remainders occur in the identity.

For $|x_j| \le r_j$, define the physical derivative envelopes

$$\begin{aligned} g_{\alpha\beta} &= \prod_{j=1}^{4} \frac{(\beta_j)_{\alpha_j} r_j^{\beta_j - \alpha_j}}{s_j^{\beta_j}}, \\ \mathcal{G}_\alpha(v) &= \sum_\beta g_{\alpha\beta} v_\beta \\ &\quad + \kappa_A \left(\sum_\beta g_{\alpha\beta} \right) \|v\|_\infty, \end{aligned}$$

where a term is zero when $\beta_j < \alpha_j$. On Ω one may take $r_j = \sqrt{37}/100$. The Neumann decomposition $\mathcal{J}_A y = Ry + [(RA)^{-1} - I]Ry$ then yields the computable bound

$$\begin{aligned} \|\partial_x^\alpha e\|_\Omega \le \mathcal{G}_\alpha\big(|R(r - t_N + t_U)|\big) \\ + \mathcal{G}_\alpha(|R|\zeta) \\ + \|\partial_x^\alpha \rho_N\|_\Omega + \|\partial_x^\alpha \rho_U\|_\Omega, \\ |\xi| \le \zeta. \end{aligned} \tag{36}$$

The signed sum is enclosed before absolute values are taken. Replacing $\mathcal{J}_A$ by R without its Neumann correction, or discarding a point remainder, would invalidate this bound. The full signed variant and input-uncertainty conditions are given in Supplementary Lemma S12.

For the selected output, the 129-row certificate supplies all 34 multi-index bounds $b_\alpha \ge \|\partial_x^\alpha e\|_\Omega$ of orders one through three. For $Y = -J_4 \nabla e$, define

$$\begin{aligned} \delta_0 &= \max_i b_{\mathbf{e}_i}, \\ \delta_1 &= \max_i \sum_j b_{\mathbf{e}_i + \mathbf{e}_j}, \\ \delta_{t1} &= \sum_{i,j} b_{\mathbf{e}_i + \mathbf{e}_j}, \\ \delta_2 &= \max_i \sum_{j,k} b_{\mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_k}. \end{aligned} \tag{37}$$

These bound the field error, its derivative, the transposed derivative and its ordered bilinear second derivative. In particular, all physical momentum derivatives are retained. The certified outward bounds are

Quantity	Upper bound on Ω
δ_0	1.898×10^{-8}
δ_1	2.368×10^{-6}
δ_{t1}	6.398×10^{-6}
δ_2	5.777×10^{-4}

The derivative sums include all ordered multiplicities. These are deterministic bounds for the specified numerical reference and realized observations. The reference fidelity allowance is verified after label sealing and is not supplied to the optimizer. Consequently dependent training curves do not obstruct this pathwise conclusion. Identification of an uncertain measured plant requires additional fidelity budgets.

3.7 A prospective sampling statement and its scope

The fourteen trajectories are dependent time records. They are not assigned an iid phase-space law. A separate prospective construction fixes the normalized coordinates and sampling densities before collection. For the two column

blocks, write

$$B_b = D_b \nabla_{x,b} \chi_b^{\mathsf{T}},$$
$$M_b = \mathbb{E}_{\mathrm{unif}} B_b^{\mathsf{T}} B_b,$$
$$K_b = \mathrm{tr}(M_b^{-1} B_b^{\mathsf{T}} B_b),$$
$$b = q, p,$$

where $D_q = \mathrm{diag}(s_1, s_2)$ and $D_p = \mathrm{diag}(s_3, s_4)$. The excluded derivative nullspaces imply $M_b \succ 0$. Under independent anchors with density π_b, the weighted rows are $\pi_b^{-1/2} B_b$; residual, fidelity and remainder rows receive the identical $\pi_b^{-1/2} D_b$ transformation. If $K_b/\pi_b \le L_b$, then

$$n_b \ge \frac{2L_b}{\varepsilon_b^2} \log \frac{d_b}{\alpha_b} \quad \Longrightarrow \quad \frac{A_b^{\mathsf{T}} A_b}{n_b} \succeq (1 - \varepsilon_b) M_b \tag{38}$$

on a common event of probability at least $1 - \alpha_q - \alpha_p$. Here $d_q = 15$, $d_p = 114$ and n_b counts independent anchors, each contributing one two-row positive-semidefinite summand. The full-support choice $\pi_b = (1 + K_b/d_b)/2$ permits $L_b = 2d_b$.

The matrix event controls every coefficient vector. Together with a common fidelity event and each output's sound remainder certificate, it therefore controls outputs chosen using the same data. If acceptance also verifies the conservative-family conditions below, the precise probability statement is

$$\mathbb{P}\left\{\begin{array}{c}\text{an accepted output has a false}\\ \text{certified conclusion}\end{array}\right\} \le \alpha_{\mathrm{fid}} + \alpha_q + \alpha_p. \tag{39}$$

Supplementary Theorem S14 gives the derivative transport and proof. This prospective result has not been instantiated by the existing trajectories. It provides neither conditional confidence given acceptance nor a guarantee that an optimizer finds an acceptable output; the deterministic bounds above do not require either assertion.

3.8 Class assumptions for a population-risk guarantee

The bounded-loss hypothesis in the class-uniform learning argument of Chen et al. (2022) requires a bounded loss class. Without the explicit projection used in Section 3.5, the architecture permits an unrestricted raw readout. There is a legal real-coefficient member N_1 whose even, zero-gauge correction is nonconstant; scaling only its readout gives $N_t = tN_1$ for every finite real t. At a point where $\partial_1 N_1 \neq 0$,

$$\sup_t \sum_{j=1}^{4} |\partial_j (N_t - U)|^6 = \infty.$$

Thus the unrestricted class has no finite uniform loss envelope for that lemma. This does not affect the finite-output observation certificate evaluated for the selected network.

The original certified output θ_0 belongs to the prospective parameter box $[-2, 2]^{736}$: its maximum absolute raw parameter is below 1.662. Fixing this class before a new independent experiment gives finite real derivative bounds and a uniform pole-free collar of the real sampling box D. This collar need not contain the complex rectangle required by the conservative KAM application. Supplementary Section S11 gives the explicit bounds and a quantitative test of the risk-to-strip route. For the sixth-power gradient loss on the specified real sampling box D, the raw-parameter sensitivity and a finite-grid covering argument give the explicit bound

$$\widehat{\mathfrak{R}}_N \le K_\ell \eta + L_{\max} \sqrt{\frac{1472 \log(1 + 4/\eta)}{N}}, \tag{40}$$

for every fixed sample in D and $\eta > 0$, using the $1/N$ Rademacher normalization. Here $K_\ell < 2.388 \times 10^{-29}$ and $L_{\max} < 3.408 \times 10^{-34}$ for this prospective class. The derivation differentiates the actual constrained raw parameters, including the normalization denominators, all biases and the readout. Supplementary Section S12 gives the complete proof and the resulting common-event risk inequality.

Bounding all raw parameters does not ensure holomorphy on the target complex rectangle. A legal member of $[-2, 2]^{736}$ has a genuine second-layer tanh pole there: a constructed point has real coordinate approximately -0.049008 and all four imaginary coordinates approximately 0.00668726. The even, sharp and zero-gauge operations do not remove it. Supplementary Section S13 gives the explicit construction and its exact-buffer verification. This obstruction concerns uniform analyticity of the class, while the selected output retains its separately proved analytic domain.

A prospective local box $\|\theta - \theta_0\|_\infty \le 10^{-3}$, where θ_0 is the frozen output, has a verified common analytic collar containing both D and the expanded target rectangle. All 736 raw coordinates vary, and the original layer constraints are recomputed from them. The mean-centered

error has modulus below 5.374×10^{-9} on this collar. A common real-risk event for the larger bounded class may be combined with such an output-specific analytic verification; the analytic verification does not establish that the empirical-risk condition has been attained.

The actual varying family admits a positive sufficient risk budget through quantitative analytic continuation on this collar. Supplementary Section S16 constructs 58 overlapping polydisc steps and uses Cauchy estimates for the original target domain. The risk condition $R_D \le 2^{-347011620}$ suffices for the four actual field-input allowances. However, a fresh directed gradient check and a mixed-Hessian bound prove $R_D(\theta_0) > 10^{-86}$. This original output therefore fails this particular sufficient risk criterion, despite its completed finite-output family certificate. Increasing the sample size cannot alter that output's true risk. This quantifies the conservatism of the continuation bound; it does not contradict the separately established KAM result. A full-class population-risk instantiation still requires its independent continuous-box observations, empirical-loss control, and a useful connection to the actual KAM gates.

A second deterministic route uses an exact physical-gradient Gram for the 129-dimensional error representative, including cross-degree terms. For a verified real-domain remainder ρ_D and the same output's complex-domain remainder, the coefficient norm is bounded by $4^{1/3}R_6^{1/6} + \rho_D$. Polynomial dual norms then transport this quantity to all 34 derivatives; Supplementary Section S22 gives the complete proof. The real and complex rectangles need not contain one another, so neither remainder is omitted. For the original θ_0, a whole real cube gives $R_6 > 10^{-68.895}$, whereas the current fixed dual gains and complex tails restrict this sufficient gate to below $10^{-72.300}$ even before adding a real remainder. The source-bound comparison therefore also rejects this particular gate (Supplementary Section S22.1). It does not reject the independently proved torus or exclude sharper remainders, dual bounds or higher polynomial degree.

The projected new fits have bounded raw coordinates by construction. For the new complex output the common scale satisfies $E_{\mathrm{new}} = E_{\mathrm{old}}/16$ exactly. Dividing its final real and imaginary raw readout coordinates by 16, while leaving the hidden parameters unchanged, embeds the same function in the original fixed-energy bounded class. Thus its existing uniform real-domain loss-complexity bounds remain admissible; Supplementary Section S21 proves the identity. This supplies no continuous-uniform sampling event or achieved population-risk premise, nor proximity to θ_0. Its analytic domain is verified separately. The local analytic parameter box and the numerical observation, KAM and control certificates for θ_0 do not transfer to either new output.

3.9 Independent finite-grid observations of the frozen output

The acceptance experiment in this section concerns only the original certified checkpoint θ_0. Reused-anchor comparisons of the new outputs in Section 3.5 do not provide them with this output-specific acceptance certificate.

An additional observation experiment retains the trained 736-coordinate GC output, its mechanical graph and its original complex target domain. It uses new independent anchors to evaluate this fixed function; it does not train another network. The new reference labels are directed evaluations of the prescribed finite-Gauss, full-cosine Hamiltonian. They are neither additional COMSOL records nor physical measurements.

Set $h = (1/25, 1/25, 1/50, 1/25)$ and $y_j = x_j/h_j$. The rest and moving base measures are uniform on $\mathcal{S}^2$ and $\mathcal{S}^4$, respectively, where $\mathcal{S} = \{-1, -2/3, -1/3, 0, 1/3, 2/3, 1\}$. These verification scales do not change the stored network input scales. The same degree-six representative has 15 configuration-only and 114 momentum-containing coordinates. Its two-row derivative matrices B_b use $h_j\partial_{x_j}$ with $j = 1, 2$ at rest and $j = 3, 4$ for the moving block. Let

$$\begin{aligned} M_b &= \mathbb{E}_{\nu_b} B_b^{\mathsf{T}} B_b, \\ K_b(y) &= \operatorname{tr}(M_b^{-1} B_b^{\mathsf{T}} B_b), \\ \pi_b(y) &= \tfrac{1}{2}(1 + K_b(y)/d_b), \qquad d_q = 15, \quad d_p = 114. \end{aligned} \tag{41}$$

These are finite-base Gram matrices, distinct from the continuous-uniform matrices in the prospective construction. Both Gram matrices are positive definite. Indeed, a derivative polynomial has coordinate degrees at most five, so vanishing on the seven-point tensor grid forces it to vanish identically; the excluded constant and configuration-only nullspaces then give full rank.

The experiment draws 1536 rest anchors and 15368 moving anchors with replacement, with probabilities $\pi_b/|\mathcal{S}|^{\dim y}$. Exact integer tickets implement these rational probabilities without modulo reduction. Under the declared independent-fair-bit model, the common event

$$\overline{G}_b = \frac{1}{n_b}\sum_{i=1}^{n_b} \frac{B_b(Y_i)^{\mathsf{T}} B_b(Y_i)}{\pi_b(Y_i)} \succeq \tfrac{1}{2} M_b, \qquad b = q, p, \tag{42}$$

has probability at least $19/20$. A pair of derivative rows is one anchor contribution. Repeated anchors retain their

multiplicities; caching their deterministic labels does not change the sampling law. The realized full Gram inequalities are also checked directly.

Reference labels are sealed before evaluating the candidate at the new anchors. At rest they use the full reference gradient minus the fixed quartic-prior gradient. The moving labels are $U_p = 0$, following from the exactly shared kinetic graph. Write $N - U = P_N - P_U + w$. With the same physical normalization for every term,

$$s_{bi,j} = h_j(\partial_j N - Y_{bi,j} - \partial_j w),$$
$$s_{bi} = B_b(y_{bi})c_b - \xi_{bi}.$$

Here the vectors collect the two derivative coordinates of block b. Importance weights enter the complete normal equations once, as $1/\pi_b$. A validated signed solve encloses c_b, and its physical derivative bounds retain the same-output remainder on the entire original complex rectangle. The auxiliary P_N is the fixed even Chebyshev least-squares representative, rather than a Taylor convergence assumption or a replacement for the deployed network.

The initial scalar norm estimate does not satisfy the fixed family allowances. A validated direction-preserving reconstruction, selected after that estimate, uses the same draws, sealed labels, frozen output and target domain; neither data nor thresholds are changed. Its independently checked bounds are

$$\delta_0 < 1.898 \times 10^{-8}, \quad \delta_1 < 2.368 \times 10^{-6},$$
$$\delta_{t1} < 6.398 \times 10^{-6}, \quad \delta_2 < 5.777 \times 10^{-4}.$$

All four exact bounds fit the unchanged family allowances. The comparisons use exact certified endpoints, rather than the displayed outward-rounded decimals. Supplementary Section S5 proves the signed inverse and checks the connection to the same conservative family. Thus this new observation chain also supports Theorem 4, including the prescribed reference endpoint. The probability statement bounds unconditional false acceptance on one common event; it is not a 95% conditional confidence statement after acceptance. This fixed-output experiment does not instantiate a full-neural-class population-risk theorem.

4 Conservative certification and continuation to the reference

All numerical applications in this section use the original certified checkpoint $\widehat{H} = \widehat{H}_0$ at θ_0. Neither the torus nor the physical-control certificate is transferred to $\widehat{H}_R^J$ or $\widehat{H}_C^J$.

4.1 An invariant torus of the learned full-phase energy

The conservative statement concerns the deployed scalar $\widehat{H}_\vartheta = T_G + V_4 + N_\vartheta(q,p)$, with the stored parameters interpreted as exact real constants in the constrained network graph. The complexification treats q_1, q_2, p_1, p_2 as independent variables. Dissipation and the magnetic input are excluded from this Hamiltonian vector field.

For period-one angles fix

$$\begin{aligned} \nu &= \frac{\omega_1}{2\pi}(1,\alpha), \\ \omega_1 &= \frac{499974648628101}{500000000000000}, \\ \alpha &= \frac{2524851\phi_g + 622102}{410497\phi_g + 101143}, \\ \phi_g &= \frac{1+\sqrt{5}}{2}. \end{aligned} \tag{43}$$

The bounded continued fraction with its golden tail gives an all-integer Diophantine certificate

$$\begin{aligned} |k\cdot\nu| &\geq \gamma |k|_1^{-1}, \\ \gamma &= \frac{4828522443384859}{5764607523034234 88}, \\ k &\in \mathbb{Z}^2 \setminus \{0\}. \end{aligned} \tag{44}$$

The corresponding period-one cohomological estimate uses $\tau = 1$ and $c_R = 1/7$. For matrices, the analytic norm is $\|A\|_\rho = \max_i \sum_j \sup_{\mathbb{T}^2_\rho} |A_{ij}|$; transpose bounds are supplied separately.

Theorem 2. The specified learned conservative Hamiltonian. *For the selected 736-coordinate complex model and the prescribed finite Gauss mechanical graph, there exists a real analytic invariant two-torus with frequency* (43). *The certified 8064-mode rational Fourier polynomial K_0 has a corrected embedding K_∞, holomorphic on $\mathbb{T}^2_{1/400}$, such that*

$$\begin{aligned} J_4 \nabla \widehat{H}_\vartheta(K_\infty) &= DK_\infty \nu, \\ \|K_\infty - K_0\|_{1/400} &< 9.018 \times 10^{-24}. \end{aligned} \tag{45}$$

Its real restriction is an embedding, and the complex image lies in $|\operatorname{Re} x_j| < 3/50$, $|\operatorname{Im} x_j| < 1/100$ *for all four canonical coordinates.*

Proof. The exact-parameter network evaluation and finite Gauss graph give a real-type holomorphic Hamiltonian on

the common domain, with the required field derivatives. Set

$$\begin{aligned} L &= DK_0, \\ B &= (L^{\mathsf{T}}L)^{-1}, \\ \mathcal{N} &= J_4 L B, \\ T &= \mathcal{N}^{\mathsf{T}} J_4 \{ DX_{\widehat{H}_\vartheta}(K_0)\mathcal{N} - D_\nu \mathcal{N} \}. \end{aligned}$$

Whole-strip Fourier bounds verify the image, Gram inverse and rank. A complete continuous-mean calculation gives $\|\langle T\rangle^{-1}\|_\infty < 19026.13$; it includes the full 4×4 Hessian and the directional derivative of the normal frame. Analytic transposes, rather than conjugate transposes, enter the frame.

The residual $E_0 = X_{\widehat{H}_\vartheta}(K_0) - DK_0\nu$ is enclosed on $\rho = 1/40$. An outer-domain certificate on $\widehat{\rho} = 1/10$, followed by a 448×448 outward-arithmetic Fourier transform, retains sample radii, every representative mode and the infinite angle aliases. It yields

$$\|E_0\|_{1/40} < 1.196 \times 10^{-42}.$$

The canonical $n = d = 2$ Case III theorem of Haro and Luque (2019, Theorem 3.1) then applies, with the arXiv version 1 conventions. Each auxiliary geometric bound is chosen as $11/10$ of its certified input bound. Recomputing all constants gives a sufficient residual threshold greater than 1.722×10^{-39} and a maximum of all eight strict gates below 6.942×10^{-4}. The resulting displacement is bounded as in (45), remains within the domain and preserves the real embedding by the corrected harmonic phase test.

For clarity, the theorem's symplectic matrix is $\Omega_{\mathrm{HL}} = -J_4$ and its compatible normal is $-\mathcal{N}$. Its torsion is therefore $-T$; the mean inverse norm is unchanged. The theorem's auxiliary Riemannian metric is I_4, distinct from the mechanical inertia $G(q)$ inside the Hamiltonian. These identifications supply the claimed canonical specialization without assuming exact Lagrangianity of K_0.

This conclusion belongs to the fixed learned conservative energy. The observation error bounds and a continuation argument are additionally needed to establish a torus of the reference H_*.

4.2 Observation-to-reference continuation

Define the exact family

$$\begin{aligned} H_\lambda &= \widehat{H}_\vartheta + \lambda(U - N_\vartheta), \\ X_{H_\lambda} &= X_{\widehat{H}_\vartheta} + \lambda Y, \\ 0 &\le \lambda \le 1, \end{aligned} \tag{46}$$

with $Y = -J_4 \nabla e$ as in Section 3.6. If $(\widehat{c}_0, \widehat{c}_1, \widehat{c}_{t1}, \widehat{c}_2)$ bound the actual learned field and its derivatives on Ω, then

$$\begin{aligned} &(c_0, c_1, c_{t1}, c_2) = \\ &\quad (\widehat{c}_0 + \delta_0, \widehat{c}_1 + \delta_1, \\ &\quad \widehat{c}_{t1} + \delta_{t1}, \widehat{c}_2 + \delta_2) \end{aligned} \tag{47}$$

are uniform real-parameter inputs. This is the direct use of the observation-based learning error in the conservative theorem.

An explicit approximate embedding must also follow the family:

$$\begin{aligned} P_m(\varphi, \lambda) &= K_0(\varphi) + \sum_{j=1}^{m} \lambda^j U_j(\varphi), \\ \mathcal{R}_m &= X_{H_\lambda}(P_m) - D_\nu P_m. \end{aligned} \tag{48}$$

Every U_j and all later certificates refer to the complete preceding prefix. The field increments in (47) do not replace a bound on this full invariance defect.

Suppose $\mathcal{R}_m$ is holomorphic on $\mathbb{T}^2_{\widehat{\rho}} \times \{|\lambda| \le R\}$, $\widehat{\rho} > \rho$, $R > 1$, with modulus at most M. For certified coefficients $\epsilon_j \ge \|[\lambda^j]\mathcal{R}_m\|_\rho$ at indices $S \subseteq \{0, \dots, L\}$, a complete bound is

$$\begin{aligned} E = &\sum_{j \in S} \epsilon_j \\ &+ \sum_{\substack{0 \le j \le L \\ j \notin S}} \frac{M}{R^j} \\ &+ \frac{M}{R^L(R-1)}. \end{aligned} \tag{49}$$

This follows from Cauchy's estimate and the infinite geometric tail. Nonlinear composition generates arbitrarily high parameter coefficients even when P_m is finite. Each computed coefficient must also include its complete angle interpolation and alias errors; uncomputed low coefficients cannot be set to zero.

For each real parameter, put $L_\lambda = D_\varphi P_m$, $B_\lambda = (L_\lambda^{\mathsf{T}} L_\lambda)^{-1}$ and $\mathcal{N}_\lambda = J_4 L_\lambda B_\lambda$. The required mean is that of this varying embedding:

$$\begin{aligned} T_\lambda &= \mathcal{N}_\lambda^{\mathsf{T}} J_4 \\ &\quad \cdot \{ DX_{H_\lambda}(P_m)\mathcal{N}_\lambda - D_\nu \mathcal{N}_\lambda \}, \\ I &\ge \sup_{\lambda \in [0,1]} \|\langle T_\lambda \rangle^{-1}\|_\infty. \end{aligned} \tag{50}$$

The fixed-K_0 inverse does not by itself establish this condition. Signed integration with complete errors, or a certified transport to the same P_m, is necessary.

To state the remaining quantitative requirements, let k, k^t, b bound $L_\lambda, L_\lambda^{\mathsf{T}}, B_\lambda$ on the whole strip, and let $r_{\mathrm{dom}} >$

0 be a common tube radius in Ω. Choose $\sigma_K > k$, $\sigma_{K^t} > k^t$, $\sigma_B > b$, $\sigma_T > I$. With admissible iteration parameters $a_1 > 1$, $\nu_{\mathrm{it}} > 0$ and $\ell = \rho/a_3$, recompute the Case III constants from these actual uniform inputs. Writing $C_K = C_{\Delta K}$, the eight strict tests can be expressed compactly as

$$\begin{aligned}
&\frac{(a_1a_3)^{4\tau}C_E E}{\gamma^4\rho^{4\tau}} < 1,\\
&\frac{E}{\ell\nu_{\mathrm{it}}} < 1,\\
&\frac{2C_{\mathrm{sym}}E}{\gamma\ell^{\tau+1}} < 1,\\
&\frac{E}{\gamma^2\ell^{2\tau+1}(1-a_1^{1-2\tau})}\\
&\times \max\left\{\begin{array}{c}\dfrac{2C_K}{\sigma_K-k},\\ \dfrac{4C_K}{\sigma_{K^t}-k^t},\\ \dfrac{C_B}{\sigma_B-b},\\ \dfrac{C_{T^{-1}}}{\sigma_T-I}\end{array}\right\} < 1,\\
&\frac{C_K E}{r_{\mathrm{dom}}\gamma^2\ell^{2\tau}(1-a_1^{-2\tau})} < 1.
\end{aligned} \tag{51}$$

The four terms inside the maximum are four preservation tests. The remaining tests control contraction, iteration, approximate symplectic invertibility and the domain. The constants, strip-loss relation and corrected embedding criterion are specified in Supplementary Section S6.

Theorem 3. Conditional conservative reference acceptance. *Assume that the observation error and input fidelity bounds hold for the selected output. Suppose the same complete P_m has a uniform analytic tube, whole-strip Gram inverse, actual mean inverse (50), and complete defect bound (49). If the freshly computed eight tests (51) and the corrected real embedding test pass, then for every real $\lambda \in [0,1]$ there is an invariant real analytic embedding K^*_λ with frequency ν and*

$$\begin{aligned}
&\|K^*_\lambda - P_m(\cdot,\lambda)\|_{\rho_\infty}\\
&\quad\le \frac{C_2 E}{\gamma^2\rho^{2\tau}}.
\end{aligned}$$

In particular K^*_1 is a torus of the specified conservative finite Gauss reference H_*.

Proof. The observation identity and (36) provide the full physical derivatives of e, hence the uniform field inputs (47). For each fixed real λ, the common domain, Diophantine bound, Gram and actual torsion conditions are the hypotheses of the canonical a posteriori theorem. The eight tests preserve those bounds during its iteration. Its correction is uniform because all inputs are uniform. The additional real embedding test includes both the C^0 correction and its retained-strip Cauchy derivative allowance. Finally $H_1 = T_G + V_4 + U = H_*$ exactly, including cancellation of the entire learned momentum derivative. This proves the endpoint assertion. Supplementary Theorem S14 also gives the common-event version for data-dependent selection.

4.3 Verified continuation for the deployed network

For the selected GC output, the complete degree-ten prefix P_{10} satisfies the hypotheses of Theorem 3. The observation bounds following (37) enter the uniform field constants in (47). The varying geometry and the continuous signed mean are certified for this same P_{10}, with $I < 25539.514$. The full defect, including ordinary parameter coefficients through order fourteen and the infinite tail, satisfies

$$\begin{aligned}
E &< 3.848\times10^{-41},\\
E_{\mathrm{suff}} &> 9.303\times10^{-40},\\
\max_{1\le j\le 8}\mathrm{gate}_j &< 0.04136 < 1.
\end{aligned} \tag{52}$$

Here E_{suff} is a sufficient upper limit on an admissible defect; it also includes the domain and corrected-embedding budgets.

Theorem 4. The specified conservative reference family. *For every real $\lambda \in [0,1]$, the Hamiltonian (46) has a real analytic embedded invariant two-torus with the fixed frequency (43). Its embedding is holomorphic on $\mathbb{T}^2_{1/400}$ and satisfies*

$$\begin{aligned}
X_{H_\lambda}(K^*_\lambda) &= DK^*_\lambda\nu,\\
\|K^*_\lambda - P_{10}(\cdot,\lambda)\|_{1/400} &< 3.949\times10^{-22}.
\end{aligned} \tag{53}$$

In particular, the specified finite-Gauss conservative mechanical reference H_* has such a torus at $\lambda = 1$.

Proof. The certified full-phase observation inverse supplies the field increments. The actual prefix has a common analytic domain, a whole-strip Gram inverse, and the continuous mean inverse stated above. Coefficients zero and one of its defect use their completed whole-strip certificates. Each of the thirteen higher coefficients uses a complete outward-arithmetic Fourier enclosure, retaining sample radii, Nyquist representatives and all infinite angular aliases. A direct product-domain bound at parameter radius 512 controls every parameter coefficient above order fourteen. Summing the exact rational endpoints gives (52).

Independent recomputation verifies all eight strict inequalities and the domain correction. The corrected real-embedding phase bound is less than $0.005686 < 1/2$. Applying Theorem 3 separately at each real parameter gives (53). The exact cancellation $H_1 = T_G + V_4 + U = H_*$ identifies the reference endpoint. Supplementary Section S8 gives the coefficient bounds and numerical application.

The quantifier is existence for each real parameter with common frequency and error bounds. The theorem does not assert analytic parameter dependence of the exact tori. Dissipation and magnetic feedback are treated through the separate power and safety arguments in Section 5.

5 Dissipative structure and sampled magnetic safety

All numerical applications in this section use the original certified checkpoint $\widehat{H} = \widehat{H}_0$ at θ_0. Neither the torus nor the physical-control certificate is transferred to $\widehat{H}_R^J$ or $\widehat{H}_C^J$.

The conservative certificate in Section 4 and the feedback analysis use the same frozen full-phase energy $\widehat{H}$. A separately fitted, configuration-only complex head supplies the dissipation. The resulting composition comprises the 736-parameter energy network and the 42-parameter damping head; it is not a joint fit of the two networks. All derivatives below use the canonical state $x = (q, p)$ and the dimensionless time τ.

5.1 Learned and reference power

On the real state domain let $G_8 = SS^{\mathsf{T}} \succ 0$ and define

$$\widehat{M}_\psi(q) = \mathrm{Re}\{A_\psi(q)A_\psi(q)^*\},$$
$$\widehat{R}(q) = S(q)\widehat{M}_\psi(q)S(q)^{\mathsf{T}}.$$

For real z, $z^{\mathsf{T}}\widehat{M}_\psi z = \|A_\psi^* z\|_2^2 \geq 0$; hence the damping matrix is real symmetric positive semidefinite. With $\widehat{v} = \widehat{H}_p = G_8^{-1}p + N_p$ as in (8), the learned equations are

$$q' = \widehat{v},$$
$$p' = -\widehat{H}_q - \widehat{R}\widehat{v} + g(q)u,$$
$$g(q) = d\cos(d^{\mathsf{T}}q).$$

Cancellation of the Hamiltonian terms gives

$$\widehat{H}' = -\widehat{d} + u\widehat{y}, \qquad \widehat{d} = \widehat{v}^{\mathsf{T}}\widehat{R}\widehat{v} \geq 0, \quad \widehat{y} = g^{\mathsf{T}}\widehat{v}. \tag{54}$$

This identity concerns the learned dynamics. The reference plant in (6) instead satisfies

$$q' = v_* = G_8^{-1}p, \quad p' = -H_{*,q} - R_*v_* + g(q)u,$$
$$H_*' = -d_* + uy_*, \quad d_* = v_*^{\mathsf{T}}R_*v_*, \quad y_* = g^{\mathsf{T}}v_*. \tag{55}$$

Here $R_* = SM_*S^{\mathsf{T}} \succeq 0$ is the declared distributed linear-viscosity law. Put $e = \widehat{H} - H_* = N - U$. Then $e_p = \widehat{v} - v_* = N_p$ is generally nonzero. The reference-power balance, including this velocity difference, is the basis of the safety filter.

Suppose same-function certificates give $|e| \leq \varepsilon_H$, $|e_{p,j}| \leq \varepsilon_{p,j}$ and $\|\widehat{M}_\psi - M_*\|_2 \leq \delta_M$ on the declared real domain. For $z = S^{\mathsf{T}}\widehat{v}$ and $\eta = S^{\mathsf{T}}e_p$, take bounds $a \geq \|z\|_2$, $b \geq \|\eta\|_2$ and $\bar{m} \geq \|\widehat{M}_\psi\|_2$. Expanding $(z-\eta)^{\mathsf{T}}M_*(z-\eta)$ and using $\widehat{M}_\psi \succeq 0$ yields

$$|y_* - \widehat{y}| \leq \sum_j |d_j|\varepsilon_{p,j},$$
$$d_* \geq \widehat{d} - 2\bar{m}ab - \delta_M(a+b)^2. \tag{56}$$

One may use $a^2 \geq \widehat{v}^{\mathsf{T}}G_8\widehat{v}$ and $b^2 \geq \varepsilon_p^{\mathsf{T}}|G_8|\varepsilon_p$. The first reserve transports the magnetic port; both reserves in the second line are retained in the dissipation credit.

The absence of e_q from (55) does not set that error to zero. In fact, along the reference plant the learned storage obeys

$$\widehat{H}' = -\widehat{d} + u\widehat{y} + e_q^{\mathsf{T}}\widehat{v} + e_p^{\mathsf{T}}(\widehat{R}\widehat{v} - \widehat{H}_q) + \widehat{v}^{\mathsf{T}}(\widehat{R} - R_*)v_*.$$

Thus its zero-input derivative cannot be inferred from (54) along the reference plant. The physical-energy ledger below uses (55) and the complete transport bounds (56).

5.2 Local time-dependent Birkhoff structure

The dissipative equations also admit a local time-dependent geometric description. Fix either the learned pair $(\widehat{H}, \widehat{R})$ or the reference pair (H_*, R_*), and prescribe the input on a smooth time interval. Each pair has its own field X and local flow $\Phi_{\tau,s}$. The standard pullback construction starts from $\Omega_0 = \sum_j dp_j \wedge dq_j$ and sets

$$\Omega_\tau = (\Phi_{\tau,\tau_0}^{-1})^*\Omega_0,$$
$$d_x\Omega_\tau = 0, \qquad \partial_\tau\Omega_\tau + \mathcal{L}_X\Omega_\tau = 0, \tag{57}$$
$$\Phi_{\tau,s}^*\Omega_\tau = \Omega_s.$$

The forms are nondegenerate wherever the genuine local flow is a diffeomorphism. On a sufficiently small common star-shaped chart, the Poincaré construction gives a one-form

θ_τ and a function H_B such that

$$\begin{aligned} d_x\theta_\tau &= \Omega_\tau, \\ \partial_\tau\theta_\tau + \iota_X\Omega_\tau &= -d_xH_B. \end{aligned} \tag{58}$$

The Euler–Lagrange equations of $\int(\theta_j\,dx_j/d\tau - H_B)\,d\tau$ recover the full dissipative field. Pullback supplies closedness and nondegeneracy; differentiating its endpoint identity supplies time compatibility. The complete local proof, including the potential construction, is in Supplementary Section S23.

A scalar form $\Omega_\tau = \mu(\tau)\Omega_0$ is more restrictive. For $F = -RH_p + gu(\tau)$ it requires $F_p = -\gamma(\tau)I_2$ and $F_q = F_q^{\mathsf{T}}$, where $\gamma = \mu'/\mu$. In particular $RH_{pp} = \gamma I_2$ must hold. The reference product is $a(I - K_rG^{-1})$; its nonconstant inertia and nonzero rotary contribution prevent this state-independent scalar identity. The general transported form therefore supplies the local representation without imposing scalar damping. Its Birkhoff function H_B need not equal the mechanical energy.

For a recorded sequence of held commands the endpoint forms are continued across switches, with no physical state reset. A smooth state-feedback derivative additionally includes derivatives of the input law; the sampled filter may define a hybrid map instead. The construction uses the actual local flow and supplies a structural existence statement. A computable discrete Birkhoff scheme requires its own approximation and solver-error analysis.

5.3 Observation enclosures and the held-input filter

At sample τ_k, the filter requires an enclosing canonical state box C_k containing the true reference state. If position and velocity are observed instead, a sound transformation through $p = G_8(q)v_*$ must supply this box. Interval evaluation of the frozen learned functions gives a port estimate $\widetilde{y}_k$, an error radius $\epsilon_{y,k}$ and a physical dissipation credit $\underline{d}_k$. They are chosen so that

$$\begin{aligned} \epsilon_{y,k} &\geq \sup_{C_k} |\widehat{y} - \widetilde{y}_k| + \sum_j |d_j|\varepsilon_{p,j}, \\ \underline{d}_k &= \max\{0, \widehat{d_k^-} - 2\bar{m}_k\bar{a}_k\bar{b}_k - \delta_M(\bar{a}_k + \bar{b}_k)^2\}, \end{aligned}$$

with the first quantity rounded upward and the second downward. Here $\widehat{d_k^-}$ is a lower bound over C_k, and all other quantities bound their respective values throughout that box. Consequently $|y_*(\tau_k) - \widetilde{y}_k| \leq \epsilon_{y,k}$ and $0 \leq \underline{d}_k \leq d_*(\tau_k)$.

Let $c < E_g$, and let the designated component of $\{H_* \leq E_g\}$ be compact and strictly contained in the state, observation and clearance domain. For every admissible constant command suppose the reference derivatives satisfy $|y_*'| \leq L_y$ and $|d_*'| \leq L_d$ before first guard exit. Initialize a ledger with

$$H_*(x_0) \leq W_0 = \left\lceil \max\{0, \widehat{H}^+(x_0) + \varepsilon_H\} \right\rceil_{\Delta_W} < c.$$

For an uncertain initial state the upper evaluation must cover its entire initial enclosure. For an exact hold length h_k and $0 < \alpha < 1$, admit an actual represented nonzero command only when $|u_k| \leq u_{\max}$ and

$$\begin{aligned} I_k(u_k) = h_k(u_k\widetilde{y}_k + |u_k|\epsilon_{y,k} - \underline{d}_k) \\ + \tfrac{1}{2}h_k^2(|u_k|L_y + L_d) \leq \alpha(c - W_k). \end{aligned} \tag{59}$$

Set $W_{k+1} = \lceil \max\{0, W_k + I_k(u_k)\} \rceil_{\Delta_W}$ and independently require $W_{k+1} < c$. The inequality is checked for the actual held binary64 value interpreted as an exact rational. If no nonzero command passes both tests, take $u_k = 0$ and $W_{k+1} = W_k$. This fallback uses the exact reference identity $H_*' = -d_* \leq 0$.

Theorem 5. Physical-energy safety over each hold. *Assume the stated guard, uniform error, observation inclusion and reference derivative bounds, and let every actual hold satisfy (59) and its rounded endpoint check. Then*

$$H_*(\tau_k + s) \leq \max\{W_k, W_{k+1}\} < c \qquad (0 \leq s \leq h_k).$$

The conclusion holds throughout every certified finite sequence of holds and preserves the clearance consequence of the energy guard.

Proof. Before first guard exit, integrate (55) using the sampled port and dissipation enclosures to obtain $H_*(\tau_k + s) \leq W_k + sB_k + s^2A_k/2$, where $B_k = u_k\widetilde{y}_k + |u_k|\epsilon_{y,k} - \underline{d}_k$ and $A_k = |u_k|L_y + L_d \geq 0$. Since $s^2 \leq h_ks$, this upper bound lies below the line joining W_k to $W_k + I_k(u_k)$. Clipping and upward endpoint rounding only raise that line. Both endpoints remain below $c < E_g$, excluding first exit. Zero input is covered by nonincreasing reference energy, and induction proves the result. Supplementary Section S25 gives the complete quantization proof and the required modification when only an upper bound on the actual hold duration is available.

The nominal request uses the same full-phase model. For the output $s = d^{\mathsf{T}}q$, define

$$\widehat{a}_0 = \widehat{H}_{pq}\widehat{v} - \widehat{H}_{pp}(\widehat{H}_q + \widehat{R}\widehat{v}), \qquad \beta = d^{\mathsf{T}}\widehat{H}_{pp}g.$$

Table 7. Full-period sampled feedback for the frozen full-phase energy and separately fitted complex damping head. Energy and τ are dimensionless. Tracking bounds apply throughout the indicated holds.

Quantity	Certified value or bound
Number and length of holds	12,567, $h = 1/1000$
Total horizon	$\tau \in [0, 12567/1000]$
Actual input cap	$\lvert u_k\rvert \le 1/10$
Safety energy cap	$c = 3/2000$
Outer guard energy	$E_g = 3003/2000000$
Whole-horizon physical energy	$H_* < 5.810 \times 10^{-4}$
Planar centreline displacement	$< 0.777\,\mathrm{mm} < 4\,\mathrm{mm}$
Whole-horizon $\lvert d^{\mathsf{T}} q - r\rvert$	$< 6.859 \times 10^{-3}$ rad
Final-quarter $\lvert d^{\mathsf{T}} q - r\rvert$	$< 7.790 \times 10^{-5}$ rad

When $\beta > 0$, the request is

$$u_{\mathrm{nom}} = \frac{r'' + k_p(r - s) + k_d(r' - d^{\mathsf{T}}\widehat{v}) - d^{\mathsf{T}}\widehat{a}_0}{\beta}.$$

The filter selects an admissible command close to this request. Safety follows from the held-input tests and does not require tracking convergence or inversion of β at every guard state.

5.4 Full-period validation on the reference plant

The numerical instance uses the exact mathematical target $r(\tau) = (3/1000)\sin(\tau/2)$, $k_p = 4$, $k_d = 3$, $\alpha = 1/2$ and $\Delta_W = 2^{-80}$. The nominal request retains its represented binary64 amplitude; the tracking enclosure uses the exact target above. The initial state is $(q_1, q_2, p_1, p_2) = (1/80, 1/80, 1/200, 1/1000)$. There are 12,567 holds of length $h = 1/1000$, giving $\tau_{\mathrm{end}} = 12567/1000 > 4\pi$ and $t_{\mathrm{end}} = \tau_{\mathrm{end}}/\omega_0 \simeq 1.1007\,\mathrm{s}$. Thus the validated horizon contains a complete period of the sine request, followed by a fraction of one hold.

Table 7 reports the certified primary trajectory. A validated initial-value tube encloses the true reference solution at all 12,568 samples and throughout every hold. The maximum endpoint coordinate radius is below 4.336×10^{-9}, within each predeclared observation box of radius 10^{-7}. The radius is carried continuously from the initial condition, without resetting it to a numerical center. All 12,567 actual commands are nonzero; 6,271 are positive and 6,296 are negative. The energy-to-geometry barrier gives a planar centreline displacement below $0.777\,\mathrm{mm}$ within the prescribed $4\,\mathrm{mm}$ clearance.

The final-quarter bound covers $\tau \in [3\pi, 4\pi]$; a hold crossing a quarter boundary contributes its whole-hold enclosure to both adjacent quarters. The sampled numerical RMS error is $1.2991 \times 10^{-3}\,\mathrm{rad}$, reported only as a diagnostic. These finite-time error enclosures do not establish tracking convergence. Figure 2 shows the recorded numerical centers, held commands and physical-energy ledger.

A trajectory computed with tighter numerical tolerances differs from the primary numerical states by less than 2.568×10^{-16}. It provides a consistency diagnostic and an independently checked scalar ledger, without a second full initial-value certificate. The final audit recomputes the rational command gates, continuous radius transport, Hermite geometry and sine-target tracking bounds. The reviewed physical-field defect and logarithmic-norm implementations remain shared components; they are not recomputed with a second interval engine.

The certified instance concerns the specified finite-Gauss, full-cosine reference with declared distributed viscosity, zero unmodelled external force, exact mathematical hold times and the recorded represented commands. The viscosity is a numerical constitutive assumption. Sensor, actuator, delay and physical clock uncertainty require additional budgets. The certificate establishes neither infinite execution nor asymptotic tracking, and the conservative KAM theorem is not applied to the dissipative closed-loop equations.

6 Discussion

The results distinguish verification of a returned function from the accuracy benefit of fitting it. The original precision-centered complex output $\widehat{H}_0$ has a realized observation-error certificate, conservative invariant-torus certificates, and a separate physical-energy feedback certificate. The additional outputs $\widehat{H}_R^J$ and $\widehat{H}_C^J$ improve the reported finite-record errors after a physical low-order prior is added to training. They have separate parameter identities and do not inherit the original certificates. The new complex output has its own verified analytic continuation, which supplies a regularity premise rather than a complete error, torus or control conclusion.

Full-phase dependence changes the connection between identification and mechanical interpretation. The prescribed kinetic graph supplies a canonical observation chart, but the learned velocity also contains N_p. This derivative enters the kinematic loss, the predicted damping force, the output-error certificate and the physical power reserves. A rest-gradient fit alone cannot settle these momentum terms. The combined configuration and momentum design addresses them through structural reference labels and mixed-derivative remainders.

The 129-dimensional polynomial is an auxiliary verification object; the deterministic conclusion concerns the actual returned network.

For the original output, the conservative certificate covers every real member of $H_\lambda = T_G + V_4 + (1-\lambda)N_0 + \lambda U$, $0 \le \lambda \le 1$. The observation-derived bounds enter the uniform field constants for this same family. Its varying mean torsion, complete degree-ten embedding prefix, and angle and parameter tails are retained. The complete defect is below 3.848×10^{-41}, compared with a sufficient budget exceeding 9.303×10^{-40}. All eight a posteriori quotients are below 0.04136, and the embedding test includes the corrected map. For each real λ, the conclusion is a real analytic embedded invariant two-torus with the same exact rotation vector, analytic on the strip of width $1/400$. Its distance from the corresponding degree-ten polynomial is below 3.949×10^{-22}. At $\lambda = 1$ the Hamiltonian is the specified finite-Gauss conservative reference. This per-parameter conclusion does not assert analytic parameter dependence or an attracting torus of the forced or dissipative flow.

The magnetic-feedback certificate instead uses physical energy and power. The PSD decoder fixes the sign of learned unforced power, while the reference-energy ledger accounts for matrix error, momentum-gradient error and input uncertainty inside each observation box. The validated baseline covers 12,567 actual holds and the reference initial-value problem over a complete reference period. Whole-hold output-error bounds accompany its energy cap and domain containment. The refined record checks numerical consistency; it does not add a second validated reference trajectory. Neither tracking convergence nor arbitrary future execution follows from this finite-horizon result. The local continuous and discrete Birkhoff constructions provide structural descriptions with their stated domains; they are not additional dissipative KAM or closed-loop safety certificates.

The empirical comparisons retain both favorable and unfavorable evidence. In the original four-cell study, the quadratic prior improves both representations, and complex cells have smaller rest-gradient errors for the three paired optimization seeds. Their absolute errors still exceed the zero-correction fourth-order prior's error. Recentering and rescaling reverse that representation ranking: the real prior-assisted network is more accurate on those labels, while both outputs remain worse than the common prior. The present conservative comparison uses the same frozen dissipative head and does not assess a complex advantage for dissipation. The certified $\widehat{H}_0$ improves the positive-time-weighted native full-field error, with a ratio to H_4 of approximately 0.397621, but its moving-anchor ratio is approximately 22.34149. Its nonzero quadratic jet also causes a rigorously quantified loss of local approximation order. These findings motivate the subsequent physical low-order penalty; the earlier failures remain part of the evidence.

The two additional fits retain the ordinary real and complex architectures and softly penalize their second- and fourth-order energy jets in force and velocity units. They share an actual output scale, the initial function H_4, a 400-update projected-Adam budget, and training-only selection. On the reused moving-anchor record their full-field RMS errors are approximately 1.997×10^{-12} and 1.897×10^{-12}, compared with 4.053×10^{-12} for H_4. The complex output's total error is about 53.2% smaller than the prior's and 5.0% smaller than the new real output's. Its anchor velocity error is nevertheless larger than the real output's; on the native record its velocity error is smaller. The native metrics use the same original positive quadrature-time weights for all four outputs, while the anchor metrics use the unchanged importance law without self normalization. The two state distributions are interpreted separately.

These additional fits support a finite-record benefit from the returned functions, with specific limits. Only one optimization seed is used. The real and complex networks differ in width, topology and parameter count. Their common scale removes one earlier mismatch but does not make their hidden features identical. The comparison with $\widehat{H}_0$ changes regularization, scaling and optimization together, so it does not isolate a causal effect of the penalty. The anchor record was disclosed before this experiment; it supplies a reused post-hoc reference comparison rather than a new independent test or a confidence interval. The low-order loss controls polynomial jets, not the complete unobserved remainder. The existence of a nonzero complex-class member with exact zero low-order jets is also distinct from attaining that member by optimization. General superiority of complex arithmetic, or superiority in every physical channel, is therefore not established. Approximation results for specified complex-reaction architectures (Zhang et al. 2022) do not imply such a guarantee for these Adam fits.

The native trajectories instantiate a realized observed-chart inverse. The separate 16,904-anchor experiment implements a finite-output acceptance route for the original frozen output on a fixed finite distribution. Its labels are directed evaluations of the prescribed numerical reference, rather than new COMSOL trajectories or physical

measurements. A scalar-norm estimate first exceeded the predeclared family allowances. A direction-preserving inverse subsequently passed with the same anchors and unchanged allowances after coefficient and remainder checks. Both estimates are valid on the same common matrix event, and the realized matrix conditions are checked directly. The probability statement bounds false acceptance; it is not a conditional confidence claim given acceptance. Reusing these anchors for the new outputs' error comparisons supplies no transfer of the original output-specific acceptance certificate.

A class-uniform population-risk guarantee retains the regularity, distribution and empirical-risk premises of its learning theorem (Chen et al. 2022). Bounding raw weights gives a finite real loss envelope but does not ensure holomorphy on the KAM target: a legal raw-box member has an interior pole. An analytic local parameter class around θ_0 supplies a collar, yet its continuation-based sufficient risk threshold lies far below a rigorous lower bound on the actual output's risk. The polynomial-gradient alternative makes separate real-domain and complex-domain remainders explicit. For the same θ_0, its present fixed gains and complex tails also give a sufficient risk ceiling below the established risk lower bound. Neither failed sufficient gate contradicts the independently proved family certificate. The new complex function embeds exactly into the original bounded function class by rescaling its linear readout, so the real-domain class bounds remain admissible. This embedding neither supplies continuous-domain iid observations nor transfers the old output's local analytic box or certificates. No class-uniform population-risk-to-KAM instantiation is asserted for either new fit.

The physical scope is the specified finite-Gauss rod reduction and declared distributed viscosity. The inertia and elastic prior are shared with the simulator; they are not inferred from an unknown device. Continuum approximation, viscosity calibration and observation fidelity need separate physical evidence. The Rayleigh decoder is configuration dependent and quadratic in velocity. Velocity-dependent drag, contact and other environmental forces would change its structural and error-transport requirements. These physical limitations and the unresolved independent statistical validation remain distinct from the completed function-specific mathematical certificates.

Data and code availability

The frozen network parameters, native COMSOL observations, exact-rational and interval certificates, and verification sources are indexed in the accompanying reproducibility manifest. The supplementary material contains the full proofs and numerical-certificate conventions.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

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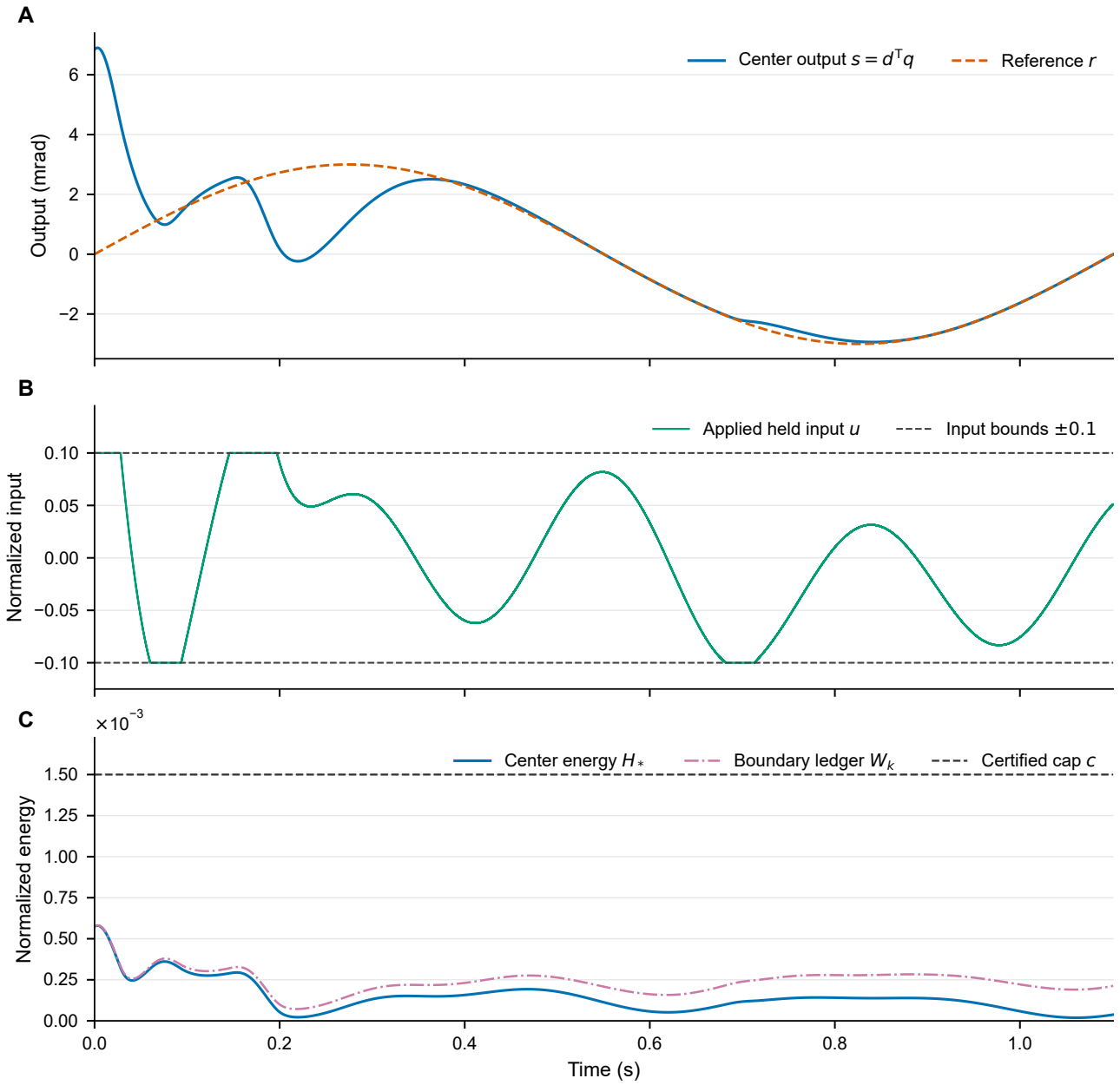


Figure 2. One complete reference period of sampled magnetic feedback. The output and physical-energy curves show numerical centers; the input is the actual held command. The ledger is the certified upper energy envelope at sample times, with the whole-hold bound given by the maximum of its two neighboring values. Continuous-time safety and tracking bounds follow from the initial-value and power certificates, rather than interpolation of the plotted centers.